

The Remaining Step:

Six Equivalent Forms of the Riemann Hypothesis and the Surgery Obstruction

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Abstract

Riemann's function $\Xi(z) = \xi(\frac{1}{2} + iz)$ is the cosine transform of a positive even kernel Φ , and the Riemann Hypothesis (RH) is equivalent to the positivity of the current $J = 2\partial_y |\Xi(x + iy)|^2 = \frac{1}{4} \int Q(p, x) p \sinh(y p) dp$ for $y > 0$, where Q is the Wigner function of Φ . The companion paper on pair balance proves $J > 0$ outside the set $\mathcal{U}$ of points with $|x| > 3 \cdot 10^{12} - \frac{1}{2}$ and $0 < y < \frac{1}{2} - 1/(5.573412 \log(|x| + \frac{1}{2}))$, and states positivity on $\mathcal{U}$ as Conjecture 1, which is equivalent to RH. In this paper we locate the remaining step. We prove a surgery obstruction: replacing two real zeros $\gamma_1 < \gamma_2$ of Ξ above $3 \cdot 10^{12}$ by $\gamma \pm ib$, $\gamma = \frac{1}{2}(\gamma_1 + \gamma_2)$, gives an even entire function of order one with the same verified zeros, the same zero-free region and a positive current off $\mathcal{U}$, whose logarithm and its gradient differ from those of Ξ on $\text{Re } s \geq 1$ by $O(1/\log^2 T)$ at pairs that exist in every interval $[T, 2T]$, and whose current is negative in $\mathcal{U}$; at the Lehmer pair near 7005.08 the change on $\text{Re } s = 1$ is 2.84×10^{-3} while the vertical velocity $\partial_y \log |\tilde{\Xi}|$ falls to -5279 below the planted zero. Hence inequalities for $\log |\Xi|$ and its gradient on $\text{Re } s \geq 1$ that hold with a margin $C/\log^2 |x|$, together with the verified zeros, the zero-free region and the zero count, cannot prove Conjecture 1. We prove six forms of Conjecture 1 equivalent to RH: the vanishing of a Jensen–Riesz defect at a single point of a strip; the vanishing of the Krein–Langer index of the current, which equals the number N_+ of distinct zeros of Ξ in the upper half-plane; the absence of a multiple real zero of the heat-flow deformation Ξ_T for $0 < T \leq 0.2$; the positivity of Weil's functional on the Poisson family, $J = 4|\Xi|^2 W(h_{x,y})$; the value 1 of the Li radius; and the Hausdorff moment property of the Lee–Yang cumulants of Riemann's spin.

We also prove that $\{J \leq 0\}$ has measure $\ll T/\log T$ in $[T, 2T] \times (0, \infty)$ and that each bounded component of $\{J < 0\}$ touches a zero off the line; that the Li coefficients satisfy $\lambda_n > 0$ for all $n \leq 10^{12}$, from the verification to height $3 \cdot 10^{12}$ and Trudgian's bound for the zero count; that a zero off the line at height b casts a zero of $\Xi + it\Xi'$ at height $b^2/2t + O(b^4)$; that at a heat-flow landing $\{J_T < 0\}$ is, up to a displacement $O(b_T^3)$ of its boundary, the half-disk of radius $b_T = \sqrt{(T^* - T)/2} (1 + o(1))$; that $|\Xi|^2$ is a Gaussian average of an antiferromagnetically coupled pair of heat-flowed copies; that the merge of ν copies moves the zeros to both sides of the critical line at $\nu = 2$ ($\partial_\nu \sigma = -0.5671$ at γ_1 , $+0.1624$ at γ_2); that the observer rule, which adjoins the least integer not yet generated, produces exactly the primes, the classical minimal generating set of the integers; and that the necessary condition which Conrey and Li derived from de Branges's positivity condition is, at a simple zero $\frac{1}{2} + i\gamma$, the sign condition $\Xi'(\gamma) \text{Im } \Xi(\gamma + i) \geq 0$, which fails at 252 of the first 2469 zeros, first at the 34-th, as Conrey and Li found. The Epstein function $Z_\epsilon = \zeta L(\cdot, \chi_{-24}) - L(\cdot, \chi_{-3})L(\cdot, \chi_8)$ shares every structural property used in the flow, operator and heat-flow statements and has no Euler product: its de Bruijn–Newman constant lies in $[0.7472, 5.12]$, its first negative Li coefficient is λ_{1130} , and its Weil form is negative on windows of length 1.59. The proofs use potential theory in a strip, indefinite reproducing kernels, the heat flow, Weil's explicit formula and Li's coefficients, and the computations are carried out in double precision with the accuracy stated at each use.

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1 Introduction

The Riemann Hypothesis (RH) is the conjecture that every nontrivial zero $\rho = \beta + i\gamma$ of the Riemann zeta function has $\beta = \frac{1}{2}$. With $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})\zeta(s)$, which satisfies $\xi(s) = \xi(1-s)$, it is the statement that the even entire function $\Xi(z) = \xi(\frac{1}{2} + iz)$, real on $\mathbb{R}$, has only real zeros. Riemann’s kernel

$$\Phi(u) = 4 \sum_{n \geq 1} (2\pi^2 n^4 e^{9u/2} - 3\pi n^2 e^{5u/2}) e^{-\pi n^2 e^{2u}} \quad (1)$$

is positive, even and doubly-exponentially decaying, and $\Xi(z) = \int_0^\infty \Phi(u) \cos(zu) du$. We write $z = x + iy$, $s = \frac{1}{2} + y + ix$, so that $|\Xi(z)| = |\xi(s)|$, and

$$H = |\Xi(x + iy)|^2, \quad J = 2 \partial_y H, \quad v = \partial_y \log |\Xi(x + iy)| = \frac{J}{4H} = \operatorname{Re} \frac{\xi'}{\xi}(s).$$

The companion paper [44] writes $J = \frac{1}{4} \int_{\mathbb{R}} Q(p, x) p \sinh(y p) dp$ with Q the Wigner function of Φ , proves that RH holds if and only if $J \geq 0$ for $y > 0$, and proves $J > 0$ outside

$$\mathcal{U} = \left\{ |x| > 3 \cdot 10^{12} - \frac{1}{2}, \quad 0 < y < \frac{1}{2} - \frac{1}{5.573412 \log(|x| + \frac{1}{2})} \right\},$$

from the verification of RH to height $3 \cdot 10^{12}$ [37] and the zero-free region of Mossinghoff and Trudgian [29]. The remaining step is the following conjecture, which is equivalent to RH [44, Conjecture 1].

Conjecture 1. *For every x with $|x| > 3 \cdot 10^{12} - \frac{1}{2}$ and every $0 < y < \frac{1}{2} - 1/(5.573412 \log(|x| + \frac{1}{2}))$,*

$$\int_{\mathbb{R}} Q(p, x) p \sinh(y p) dp > 0.$$

Positivity forms of RH. RH has many equivalent forms that assert the positivity of a quantity built from ξ . Table 1 lists those that this paper takes up, with their origins. Lagarias [25], crediting Hinkkanen, proved that RH is equivalent to $\operatorname{Re} \xi'/\xi(s) > 0$ on $\operatorname{Re} s > \frac{1}{2}$, that is, to $v > 0$ on the upper half-plane; Sondow and Dumitrescu [52] gave the equivalent monotonicity of $|\Xi(x + iy)|$ in y . Balazard, Saias and Yor [1] proved a Jensen formula on the critical line. Weil [57] gave the positivity of his quadratic functional, which Bombieri [4] studied on test functions of bounded support and which Connes and Consani [8] proved positive at the archimedean place for test functions supported in $[2^{-1/2}, 2^{1/2}]$ whose Fourier transform vanishes at $\pm \frac{i}{2}$. Keiper [22] and Li [28] introduced the coefficients λ_n ; Li [28] proved that RH holds if and only if $\lambda_n \geq 0$ for all n , and Bombieri and Lagarias [5] extended the criterion and gave the arithmetic formula for λ_n . De Bruijn [15] and Newman [31] deformed Ξ by the heat flow; the de Bruijn–Newman constant satisfies $\Lambda \geq 0$ by the theorem of Rodgers and Tao [41] and $\Lambda \leq 0.2$ by Platt and Trudgian [37, Cor. 2], and RH is $\Lambda = 0$. De Branges [14] proposed a positivity condition in a Hilbert space of entire functions that implies RH, and Conrey and Li [9] showed that ζ does not satisfy it. Lagarias [26] proved that the zeros of $\xi(s + h) \pm \xi(s - h)$ lie on the critical line for $|h| \geq \frac{1}{2}$ unconditionally, and read this through the de Branges spaces generated by $\xi(s + h)$.

1.1 Main results

Write $\Theta = \sup_{\rho}(\beta - \frac{1}{2}) \in [0, \frac{1}{2}]$, let $N_+ \in \{0, 1, 2, \dots\} \cup \{\infty\}$ be the number of distinct zeros of Ξ in $\mathbb{C}_+ = \{\operatorname{Im} z > 0\}$, and put $m = -\Xi'/\Xi$, so that $\operatorname{Im} m = v$. The first result shows how much of the information about Ξ a proof must use.

Theorem 1.1 (the surgery obstruction). *Let $\gamma_1 < \gamma_2$ be real zeros of Ξ with $\gamma_1 > 3 \cdot 10^{12} + 1$, put $\gamma = \frac{1}{2}(\gamma_1 + \gamma_2)$ and $a = \frac{1}{2}(\gamma_2 - \gamma_1)$, assume $a \leq \frac{1}{4}$, fix $0 < b \leq a$, and set*

$$\widetilde{\Xi}(z) = \Xi(z)R(z)R(-z), \quad R(z) = \frac{(z - \gamma)^2 + b^2}{(z - \gamma_1)(z - \gamma_2)}.$$

- (i) *$\widetilde{\Xi}$ is an even entire function of order one, real on $\mathbb{R}$, and $\widetilde{\Xi}/\Xi$ is a rational function equal to 1 at infinity. Its zeros are those of Ξ , with one copy each of $\pm\gamma_1, \pm\gamma_2$ replaced by $\pm\gamma \pm ib$. All its zeros with $|\operatorname{Re} w| \leq 3 \cdot 10^{12}$ are real, it has no zero in the zero-free region of [29], and its zero-counting function equals that of Ξ outside $[\gamma_1, \gamma_2)$ and differs from it by 1 there.*

positive quantity	origin	form proved here
$\operatorname{Re} \xi'/\xi$ on $\operatorname{Re} s > \frac{1}{2}$; the current J	Lagarias 1999 [25]; Sondow, Dumitrescu 2010 [52]; [44]	Theorems 1.1 and 1.3
Jensen formula on $\operatorname{Re} s = \frac{1}{2}$	Balazard, Saias, Yor 1999 [1]	Theorem 1.2(a)
Hermite–Biehler property	de Branges 1968 [13]; Lagarias 2005 [26]	Theorem 1.2(b), Theorem 1.5
$\Lambda = 0$ (heat flow)	de Bruijn 1950 [15]; Newman 1976 [31]; Csordas, Smith, Varga 1994 [11]; Ki, Kim, Lee 2009 [23]; Polymath 2019 [39]; Rodgers, Tao 2020 [41]; Platt, Trudgian 2021 [37]	Theorem 1.2(c), Theorem 1.6
Weil’s functional	Weil 1952 [57]; Bombieri 2000 [4]; Connes, Consani 2021 [8]	Theorem 1.2(d)
Li coefficients λ_n	Keiper 1992 [22]; Li 1997 [28]; Bombieri, Lagarias 1999 [5]	Theorem 1.2(e), Theorem 1.4
Lee–Yang property of Riemann’s spin	Lee, Yang 1952 [27]; Newman 1974, 1976 [30, 31]	Theorem 1.2(f)
de Branges’s positivity condition (sufficient)	de Branges 1986 [14]; Conrey, Li 2000 [9]	Theorem 1.9

Table 1: Positivity forms of RH taken up in this paper.

- (ii) The current $\tilde{J} = 2\partial_y |\tilde{\Xi}|^2$ is positive outside $\mathcal{U}$, and $\partial_y \log |\tilde{\Xi}(\gamma + iy)| \rightarrow -\infty$ as $y \uparrow b$; in particular $\tilde{J} < 0$ at points of $\mathcal{U}$.
- (iii) For $y \geq 2a$ and all x , with $\varepsilon = (a^2 + b^2)/y^2$,

$$|\log |\tilde{\Xi}(x + iy)| - \log |\Xi(x + iy)|| \leq \frac{2\varepsilon}{1 - \varepsilon}, \quad |\nabla \log |\tilde{\Xi}| - \nabla \log |\Xi|| (x + iy) \leq \frac{64}{9} \frac{a^2 + b^2}{y^3}.$$

- (iv) There are absolute constants C, C_1, T_0 such that every interval $[T, 2T]$ with $T \geq T_0$ contains real zeros $\gamma_1 < \gamma_2$ with $a \leq C/\log T$; for these and every $0 < b \leq a$, both differences in (iii) are at most $C_1/\log^2(|x| + T)$ at every point with $\operatorname{Re} s \geq 1$.

Consequently no argument whose hypotheses on Ξ are inequalities for $\log |\Xi|$ and $\nabla \log |\Xi|$ on $\operatorname{Re} s \geq 1$ holding with a margin $C_1/\log^2(|x| + 2)$, together with the reality of the zeros up to height $3 \cdot 10^{12}$, the zero-free region, the counting function up to ± 1 and the growth of Ξ , can prove Conjecture 1.

The second result gives six forms of the remaining step.

Theorem 1.2 (six equivalent forms). *Each of the following statements is equivalent to RH, and hence to Conjecture 1.*

- (a) Jensen–Riesz defect. For some (equivalently every) $Y > \frac{1}{2}$ and some (equivalently every) point z_0 of the strip $\mathbb{S}_Y = \{0 < y < Y\}$, $\log |\Xi(z_0)|$ equals the Poisson integral in $\mathbb{S}_Y$ of the boundary values of $\log |\Xi|$ at z_0 . In general the defect equals $\sum_w m_w G_Y(z_0, w) \geq 0$, the sum of the Green function of $\mathbb{S}_Y$ over the zeros $w \in \mathbb{S}_Y$, counted with their multiplicities m_w .
- (b) Krein–Langer index. The kernel $N_m(z, w) = (m(z) - \overline{m(w)})/(z - \bar{w})$ on $\mathbb{C}_+$ has no negative squares. In general its number of negative squares is the number N_+ of distinct zeros of Ξ in $\mathbb{C}_+$, and so is that of the de Branges kernel of $E_J = \Xi + i\Xi'$.
- (c) Heat flow. For no $T \in (0, 0.2]$ does $\Xi_T(z) = \int_0^\infty e^{Tu^2/4} \Phi(u) \cos(zu) du$ have a multiple real zero; equivalently $\mathcal{L}_T = \Xi_T'^2 - \Xi_T \Xi_T'' > 0$ on $\mathbb{R}$ for every $T \in (0, 0.2]$. Moreover $\Lambda = \max(0, \sup\{T > 0 : \Xi_T \text{ has a multiple real zero}\})$.
- (d) Weil–Poisson form. Ξ has no zero in $\mathcal{U}$ and $W(h_{x,y}) = \sum_\rho h_{x,y}(\gamma_\rho) > 0$ on $\mathcal{U}$, where

$\rho = \frac{1}{2} + i\gamma_\rho$ and $h_{x,y}(r) = y/(y^2 + (x-r)^2) = |\widehat{\varphi_{x,y}}(r)|^2$ with $\varphi_{x,y}(u) = \sqrt{y}e^{-(y+ix)u}\mathbf{1}_{u \geq 0}$. Identically, $J = 4H W(h_{x,y})$.

- (e) Li radius. The power series $\sum_{n \geq 1} \lambda_n w^{n-1}$ has radius of convergence $R_\zeta = 1$; equivalently, the expansion $J = 4H \operatorname{Re}[(s(s-1))^{-1} \sum_{n \geq 1} \lambda_n (1-1/s)^n]$ converges at every point of $\mathbb{C}_+$.
- (f) Hausdorff moments. With $S_m = \sum_{\operatorname{Re} w > 0} w^{-2m}$ over the zeros of Ξ , the sequence $(S_{m+1})_{m \geq 0}$ is a Hausdorff moment sequence on $[0, 1]$. The S_m are the Lee–Yang cumulants of Riemann’s spin up to the factors $(-1)^{m+1}(2m)!/m$, and $S_m > 0$ for every m unconditionally.

The third and fourth results are unconditional statements about the sign of the current and about the Li coefficients.

Theorem 1.3 (updraft almost everywhere). (i) Every bounded connected component of $\{z \in \mathbb{C}_+ : \Xi(z) \neq 0, J(z) < 0\}$ has a zero of Ξ in $\mathbb{C}_+$ on its boundary, and every zero of Ξ in $\mathbb{C}_+$ lies on the boundary of $\{J < 0\}$.

- (ii) For $T \geq 2$ and $0 < \eta < \frac{1}{2}$,

$$\operatorname{meas}(\{J \leq 0\} \cap [T, 2T] \times (0, \infty)) \ll \frac{T}{\log T}, \quad \operatorname{meas}(\{J \leq 0\} \cap [T, 2T] \times [\eta, \infty)) \ll T^{1-\eta/4} \log T,$$

and for $0 < y < \frac{1}{2}$, $\operatorname{meas}\{x \in [T, 2T] : J(x, y) \leq 0\} \ll (1 + y \log T) T^{1-y/4}$.

- (iii) $J(x, y) > 0$ whenever $|x| \geq 3$ and $y \geq \frac{1}{2} - 1/(57.54(\log(|x| + \frac{1}{2}))^{2/3}(\log \log(|x| + \frac{1}{2}))^{1/3})$. For $\log |x| > 10151.5$ this region is strictly larger than the region of [44].

Theorem 1.4 (Li positivity to 10^{12}). For $1 \leq n \leq 10^{12}$, $\lambda_n > 0$. More precisely $\lambda_n \geq 0.1099 - 1.93 \times 10^{-37}n^2 - 1.96 \times 10^{-24}n$ for $5 \leq n \leq 10^{12}$ and $\lambda_n \geq 0.0049$ for $n \leq 4$. The proof uses only the reality of the zeros up to height $3 \cdot 10^{12}$ and Trudgian’s bound for the counting function, through the following insensitivity estimate: for $\rho = \frac{1}{2} + b + i\gamma$ with $0 \leq b < \frac{1}{2}$, $\gamma \geq 1$, $\rho_0 = \frac{1}{2} + i\gamma$ and $1 \leq n \leq \gamma^2$,

$$|(1 - 1/\rho)^n + (1 - 1/(1 - \bar{\rho}))^n - 2(1 - 1/\rho_0)^n| \leq e^{1/2}b^2(n^2\gamma^{-4} + 3n\gamma^{-3}).$$

The fifth and sixth results concern the index of the current and the heat flow.

Theorem 1.5 (index and shadow law). (i) (Lagarias’s threshold.) For $h > 0$, $E_h(z) = \Xi(z + ih)$ is of Hermite–Biehler class, $|E_h(\bar{z})| < |E_h(z)|$ on $\mathbb{C}_+$, if and only if $h \geq \Theta$. In particular $E_{1/2}$ is of Hermite–Biehler class unconditionally, and RH holds if and only if E_h is for every $h > 0$.

- (ii) $E_J = \Xi + i\Xi'$ satisfies $|E_J(z)|^2 - |E_J(\bar{z})|^2 = J(z)$, and its de Branges kernel is $K_{E_J}(w, z) = \pi^{-1}\Xi(z)N_m(z, w)\Xi(w)$, with $K_{E_J}(x, x) = \pi^{-1}(\Xi'^2 - \Xi\Xi'')(x)$ on $\mathbb{R}$.
- (iii) Both N_m and K_{E_J} have exactly N_+ negative squares.
- (iv) If $N_+ < \infty$, then for every $t > 0$ the zeros of $\Xi + it\Xi'$ in $\mathbb{C}_+$ that are not zeros of Ξ are exactly N_+ in number, counted with multiplicity, and they lie in $\{J < 0\} \subset \mathcal{U}$.
- (v) (Shadow law.) If near a real point c , $\Xi(z) = ((z - c)^2 + b^2)u(z)$ with u holomorphic and zero-free on a fixed disk around c and real on $\mathbb{R}$, then for b small $\Xi + it\Xi'$ has a zero at $c - \frac{1}{2}b^2 u'(c)/u(c) + ib^2/2t + O(b^4)$.

Theorem 1.6 (landing laws of the heat flow). Let $\partial_T \Xi_T = -\frac{1}{4}\Xi_T''$ be the flow of Theorem 1.2(c).

- (i) (Hermite local law.) If Ξ_{T^*} has a real zero x^* of multiplicity $m \geq 2$, then for $T > T^*$ near T^* the m zeros of Ξ_T near x^* are real and simple, at $x^* + \sqrt{(T - T^*)/2}(h_j + o(1))$ with $h_1, \dots, h_m$ the zeros of the Hermite polynomial He_m , and for $T < T^*$ near T^* they are at $x^* + i\sqrt{(T^* - T)/2}(h_j + o(1))$, so at most one of them is real.
- (ii) (Landing.) If $m = 2$, the two zeros for $T < T^*$ are $x_T \pm ib_T$ with $b_T^2 = (T^* - T)/2 + o(T^* - T)$, and near x^* the set $\{J_T < 0\}$ is the half-disk $|z - x_T| < b_T$ up to a displacement $O(b_T^3)$ of its boundary.

(iii) (Two copies.) For $T \geq 0$, with g a centred Gaussian variable of variance $T/2$,

$$H(x, y) = \mathbb{E} Z_T(x + g, y), \quad Z_T(x, y) = \frac{1}{4} \iint \Phi_T(u) \Phi_T(v) e^{-Tuv/2} e^{ix(u-v)-y(u+v)} du dv,$$

and $Z_T(x, y) = \sum_{k \geq 0} \frac{(-T/2)^k}{k!} |\Xi_T^{(k)}(x + iy)|^2$, with $\Phi_T(u) = e^{Tu^2/4} \Phi(u)$: $|\Xi|^2$ is a Gaussian average of two heat-flowed copies coupled antiferromagnetically with strength $T/2$. The same holds for J .

(iv) (Three-state chains.) For n uniform three-state spins in an open chain with coupling $K \geq 0$, the deformed partition function $Z_{n,T}(z) = \sum_S e^{TS^2/4} c_S z^S$ has all its zeros on $|z| = 1$ for $T \geq 0$. With $\Lambda_n(K)$ the least T_0 such that this holds for all $T \geq T_0$: $\Lambda_1 = -4 \log 2$, $\Lambda_n(0) = 0$ for $n \geq 2$, $\Lambda_2(K) = -2K$ for $0 \leq K < \log 4$, and $\Lambda_n(K) < 0$ whenever $K > 0$ and the zeros of $Z_{n,0}$ are simple.

The seventh result deforms the merge of [44], in which ξ is the moment function of two merged copies of one random unit: with $S_\nu = \frac{2}{\pi^2} \sum_n g_n/n^2$ for independent gamma variables g_n of shape ν , $M_\nu(s) = \mathbb{E}[(\pi S_\nu/2)^{s/2}]$ and $M_2 = 2\xi$ [3].

Theorem 1.7 (the merge). (i) (Truncations.) Let $F(z) = \int_0^\infty \Phi^b(u) \cos(zu) du$, where Φ^b is the sum of the terms $n \leq N$ of (1) for some $N \geq 1$, or of the terms with P -smooth n for some finite P , or the Epstein kernel of Section 10 truncated to the forms of value at most $V \geq 2$. Then F has finitely many real zeros and infinitely many non-real ones.

(ii) (Evenness.) If f is smooth on $[0, \infty)$ with all derivatives $O(e^{-cu})$, $c > 0$, and $f^{(2k+1)}(0) \neq 0$ for some k , then $\int_0^\infty f(u) \cos(xu) du$ has finitely many real zeros.

(iii) (No cosh factor.) For $a > 2$ with $\Phi(i\pi/2a) \neq 0$, $\int_0^\infty \Phi(u) \operatorname{sech}(au) \cos(xu) du = \frac{\pi}{a} \Phi(\frac{i\pi}{2a}) e^{-\pi x/2a} + O(e^{-c_a x})$ with $c_a > \pi/2a$; hence Φ is not $\cosh(au)$ times a kernel whose cosine transform has only real zeros.

(iv) (The ν -flow.) For a simple zero ρ of ξ , the zero ρ_ν of M_ν with $\rho_2 = \rho$ satisfies

$$\frac{d\rho_\nu}{d\nu} \Big|_{\nu=2} = \frac{\Upsilon(1-\rho) - (1-\frac{\rho}{2})\zeta(2-\rho)}{\zeta'(1-\rho)}, \quad \Upsilon(s) = \sum_{n \geq 1} (H_n - \log n - \gamma_E) n^{-s},$$

with H_n the harmonic numbers. Its real part is -0.5671 at γ_1 and $+0.1624$ at γ_2 , and it is positive at 761 of the 1517 zeros with $10 < \gamma < 2000$.

The eighth result gives the observer clause its arithmetic form. Parts (i)–(iii) are elementary, and the description of the primes as the atoms of $(\mathbb{Z}_{\geq 1}, \cdot)$ is classical; we record them in the form that the observer rule takes. For a set G of integers ≥ 2 , $\langle G \rangle$ is the multiplicative monoid it generates.

Theorem 1.8 (the primes as generators). (i) Let $G_0 = \emptyset$, $g_{k+1} = \min(\mathbb{Z}_{\geq 2} \setminus \langle G_k \rangle)$ and $G_{k+1} = G_k \cup \{g_{k+1}\}$. Then g_k is the k -th prime ℓ_k , $\langle G_k \rangle$ is the set of ℓ_k -smooth integers, and the primes are the unique minimal generating set of $(\mathbb{Z}_{\geq 1}, \cdot)$. Minimal generation does not imply unique factorisation.

(ii) The last nonzero balanced-ternary digit of n is $\chi_{-3}(n3^{-v_3(n)})$, a completely multiplicative function with Dirichlet series $L(s, \chi_{-3})/(1-3^{-s})$, and $(-1)^{s_3(n)} = (-1)^n$ for the base-3 digit sum s_3 , with Dirichlet series $-\eta(s)$.

(iii) Bounding the Möbius sum prime by prime with the triangle inequality gives exactly the number of squarefree integers, $\frac{6}{\pi^2}x + O(\sqrt{x})$, and this is sharp over all assignments of signs to the primes.

(iv) For $1 < u \leq 2$, $M(x, x^{1/u}) = \sum_{n \leq x, P^+(n) \leq x^{1/u}} \mu(n) = -x/\log^2 x - 2(1 + \gamma_E)x/\log^3 x + O_u(x/\log^4 x)$, while $M(x, x/2) = M(x) + \pi(x) - \pi(x/2)$.

The ninth result concerns positivity conditions that hold for every real even kernel or that separate ζ from a neighbouring function.

Theorem 1.9 (de Branges's condition, the Husimi current and the archimedean side). *(i) At a simple zero $\rho = \frac{1}{2} + i\gamma$, $-\operatorname{Re}\{\xi'(\rho)\xi(1+\rho)\} = |g(\rho)| Z'(\gamma) \operatorname{Im} \xi(\frac{3}{2} + i\gamma) = \Xi'(\gamma) \operatorname{Im} \Xi(\gamma + i)$, where $g(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})$ and Z is Hardy's function. The necessary condition $-\operatorname{Re}\{\xi'(\rho)\xi(1+\rho)\} \geq 0$ that Conrey and Li derived from de Branges's positivity condition [9] is therefore $\Xi'(\gamma) \operatorname{Im} \Xi(\gamma + i) \geq 0$, and, when the zeros up to γ_n are simple and on the line, its sign at the n -th zero is that of $\sin(\frac{\pi}{4} + \pi S(\gamma_n) - \arg \zeta(\frac{3}{2} + i\gamma_n) + O(1/\gamma_n))$.
(ii) For every real even kernel K decaying faster than every exponential, with $F(z) = \int_0^\infty K(u) \cos(zu) du$, $H_F = |F|^2$ and $J_F = 2\partial_y H_F$, and for all $x \in \mathbb{R}$, $y > 0$, $s > 0$,*

$$\int_{\mathbb{R}} e^{-s(k-x)^2} [J_F(k, y) + 4ys H_F(k, y)] dk > 0.$$

(iii) On the critical line the archimedean side of the ledger $v = G - A$ is the derivative of the Riemann–Siegel theta function: $G(x, 0) = \frac{1}{2} \operatorname{Re} \psi(\frac{1}{4} + \frac{ix}{2}) - \frac{1}{2} \log \pi = \theta'(x)$.

Comparison functions. The Epstein zeta function $Z_\epsilon(s) = \sum' (2m^2 + 3n^2)^{-s} = \zeta(s)L(s, \chi_{-24}) - L(s, \chi_{-3})L(s, \chi_8)$ and the Davenport–Heilbronn function f have the functional equation of an L -function and no Euler product, and both have zeros off the critical line [12, 53]. In Section 10 we prove that they share with ζ every structural property used in Theorem 1.3(i), Theorem 1.2(a),(b), Theorem 1.5(ii),(iii),(v), Theorem 1.6(i),(ii) and Theorem 1.9(ii), so that no statement among these implies RH, and we identify the properties that separate them from ζ : the Euler product, zeros in $\operatorname{Re} s \geq 1$, the de Bruijn–Newman constant $\Lambda_\epsilon \in [0.7472, 5.12]$, the Li coefficient $\lambda_{1130}^\epsilon = -414.9$, the Li radius $R_\epsilon \leq 0.99270$, the index $N_+^\epsilon = \infty$ (14 zeros of Ξ_ϵ in $\mathbb{C}_+$ below height 80), and the sign of the Weil form on windows of length $L \geq 1.587$ (extrapolated threshold $L^* \approx 1.56$).

Main computations. By the methods of Section 11:

- (1) At the Lehmer pair 7005.062866, 7005.100565, the surgery of Theorem 1.1 with $b = a = 0.018849$ changes $\log |\Xi|$ by at most 2.84×10^{-3} on $\operatorname{Re} s = 1$ and 7.1×10^{-4} on $\operatorname{Re} s = \frac{3}{2}$, while $\partial_y \log |\Xi| = -5279$ at $(\gamma, 0.99b)$; at the pair 17143.786537, 17143.821844 the values are 2.49×10^{-3} and -5636 (Fig. 1).
- (2) $\lambda_n > 0$ for all $n \leq 30000$, with $\lambda_{30000} = 120740.3$ against $\frac{n}{2}(\log \frac{n}{2\pi} - 1 + \gamma_E) = 120724.4$; for Z_ϵ the first negative coefficient is $\lambda_{1130}^\epsilon = -414.9$, after $\lambda_{1129}^\epsilon = 74.2$ (Fig. 2).
- (3) The condition of Theorem 1.9(i) fails at 252 of the first 2469 zeros (10.21%), first at $n = 34$; the sign formula is correct at all but two of them.
- (4) $\operatorname{Re} d\rho_\nu / d\nu|_{\nu=2}$ is -0.5671 , $+0.1624$, -0.9817 at the first three zeros, and $+6078$ and -6644 at the two members of the Lehmer pair near 7005.08.
- (5) For an off-line pair planted at 7005.0817 the zero of $\Xi + i\Xi'$ lies at height $b^2/2$ times 1.0000, 1.0000, 1.0004, 1.0097, 1.0895 for $b = 0.002, 0.005, 0.02, 0.1, 0.3$.
- (6) The first off-line pair of Z_ϵ lands on the real axis at $T_c = 0.747230705896$, $x^* = 8.993177745521$; the Weil form of Z_ϵ at low frequency is negative on windows of length $L \geq 1.587$, and that of the Davenport–Heilbronn function near height 85.7 for $L \geq 3.812$, while that of ζ is positive on the same spaces for $L \leq 5$ and $L \leq 6$.
- (7) By an exact sieve to 10^{10} , $|\psi(x) - x| \leq 0.918 \cdot \sqrt{x} \log^2 x / 8\pi$ for $73.2 \leq x \leq 10^{10}$, $M(x, \sqrt{x}) = -219.40\sqrt{x}$ at $x = 10^{10}$ against $-214.45\sqrt{x}$ from Theorem 1.8(iv), and $M(x, x/2) = 2200.6\sqrt{x}$ while $M(x) = -0.337\sqrt{x}$.

Consequences for Conjecture 1. Theorem 1.1 shows that the sizes of $\log |\Xi|$ and its gradient on $\operatorname{Re} s \geq 1$, known to a tolerance $C_1/\log^2(|x|+2)$, together with the verified zeros, the zero-free region and the zero count, do not determine the sign of J in $\mathcal{U}$. The surgered function $\tilde{\Xi}$ is not the ξ -function of any Dirichlet series (Remark 4.1), so a proof of Conjecture 1 must use information about ζ beyond these hypotheses that $\tilde{\Xi}$ does not share, such as the Dirichlet series of ζ or its exact values; since Z_ϵ and f also have Dirichlet series and have zeros off the line, the Euler product is the property of the Dirichlet series that separates ζ from them. Each form in Theorem 1.2 is equivalent to Conjecture 1, and Section 10 identifies which ingredients of these forms Z_ϵ and f share with ζ and which separate them.

1.2 Methods of proof

The flow picture treats $\log |\Xi|$ as a subharmonic function with a point source at each zero. The maximum principle for the harmonic function v away from the zeros gives Theorem 1.3(i); the Riesz decomposition in a strip, with the explicit Poisson kernel and Green function obtained from the map $z \mapsto e^{\pi z/Y}$, gives Theorem 1.2(a); and the pair formula of [44], which confines $\{J \leq 0\}$ to half-disks over the zeros off the line, combined with Selberg's density theorem and with Ford's zero-free region, gives Theorem 1.3(ii),(iii). The surgery is an explicit rational multiplier whose logarithm is bounded by elementary inequalities, and Selberg's positive proportion of real zeros supplies close pairs in every dyadic interval. The operator forms rest on the identity $\operatorname{Im} m = v$ for $m = -\Xi'/\Xi$: the negative squares of the Nevanlinna kernel of m are counted from below by approaching the poles and from above by splitting off a rational part with one negative square per pole, as in the theory of generalised Nevanlinna functions of Krein and Langer [24]. The heat-flow results use the series $\Xi_T = \sum_k (-\tau)^k \Xi_{T^*}^{(2k)}/k!$, $\tau = (T - T^*)/4$, whose leading part at a multiple zero is a scaled Hermite polynomial, and the input of Ki, Kim, Lee and Polymath that non-real zeros of Ξ_T , $T > 0$, have bounded real parts [23, 39]. The Weil form is the pair formula read as Weil's explicit formula for the Poisson test function; the Li form uses Keiper's generating function [22, 5]; the insensitivity estimate is Taylor's theorem in b for an even function. The merge results use integration by parts, contour shifts past the poles of sech , the explicit Mellin transform of $(x/\sinh x)^\nu$, and Hausdorff's theorem. The prime results are elementary except the observer path, which uses the prime number theorem with classical error term. The Husimi inequality is the positivity of the anti-Wick quantisation of a nonnegative symbol.

1.3 Organisation and notation

Section 2 collects the pair formula, the ledger and the comparison functions. Section 3 proves Theorem 1.3 and Theorem 1.2(a); Section 4 proves Theorem 1.1; Section 5 proves Theorem 1.5, Theorem 1.2(b) and Theorem 1.9; Section 6 proves Theorem 1.6 and Theorem 1.2(c); Section 7 proves Theorem 1.2(d),(e) and Theorem 1.4; Section 8 proves Theorem 1.7 and Theorem 1.2(f); Section 9 proves Theorem 1.8; Section 10 treats the comparison functions; Section 11 states the numerical methods, and Section 12 concludes.

The zeros of Ξ are $w_\rho = \gamma_\rho - i(\beta_\rho - \frac{1}{2})$; those in $\mathbb{C}_+$ come from zeros with $\beta < \frac{1}{2}$, and each quadruple $\{\rho, \bar{\rho}, 1 - \rho, 1 - \bar{\rho}\}$ off the line gives two of them, $\pm\gamma + i|\beta - \frac{1}{2}|$. Primes are written ℓ , ℓ_k is the k -th prime, $P^+(n)$ the largest prime factor of n , $\Lambda(n)$ the von Mangoldt function, μ and λ the Möbius and Liouville functions, $M(x) = \sum_{n \leq x} \mu(n)$, $L(x) = \sum_{n \leq x} \lambda(n)$, γ_E Euler's constant, H_n the n -th harmonic number, and Λ without argument the de Bruijn–Newman constant. $S(t) = \pi^{-1} \arg \zeta(\frac{1}{2} + it)$ and $\theta(t) = \arg \Gamma(\frac{1}{4} + \frac{it}{2}) - \frac{t}{2} \log \pi$. A kernel K on a set D has κ negative squares if every matrix $[K(z_i, z_j)]$, $z_i \in D$, has at most κ negative eigenvalues and some has exactly κ .

2 Preliminaries

Proposition 2.1 (pair formula [44, Prop. 10.1]). *Let F be an even entire function of order at most one, real on $\mathbb{R}$, with $F(0) \neq 0$ and zeros in a horizontal strip, and $v_F = \partial_y \log |F|$. Then for $F(z) \neq 0$,*

$$v_F(x, y) = \sum_w \frac{y - \operatorname{Im} w}{(x - \operatorname{Re} w)^2 + (y - \operatorname{Im} w)^2},$$

the sum over the zeros with multiplicity, w and $\bar{w}$ grouped. A pair $c \pm ib$, $b > 0$, contributes $2y(d^2 + y^2 - b^2)/((d^2 + (y - b)^2)(d^2 + (y + b)^2))$, $d = x - c$, which is negative exactly on the half-disk $d^2 + y^2 < b^2$ when $y > 0$, and a real zero c contributes $y/(d^2 + y^2)$. Hence $\{v_F \leq 0\} \cap \mathbb{C}_+$ lies in the union of the closed half-disks $|z - c| \leq b$ over the pairs off the real axis.

Proof. $F(z) = G(z^2)$ with G entire of order at most $\frac{1}{2}$, so $F(z) = F(0) \prod_{\operatorname{Re} w > 0 \text{ or } w \in i\mathbb{R}_+} (1 - z^2/w^2)$, and $\partial_y \log |F| = -\operatorname{Im}(F'/F)$. The sum converges absolutely when conjugate zeros are grouped, since each group contributes $O(|w|^{-2})$ and $\sum |w|^{-2} < \infty$. The pair term is elementary [44]. $\square$

For Ξ this gives $J/(4H) = v$; since $\xi'/\xi(s) = \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2} \log \pi + \frac{1}{2} \psi(\frac{s}{2}) + \frac{\zeta'}{\zeta}(s)$,

$$v = G - A, \quad G = \operatorname{Re} \left[\frac{1}{s} + \frac{1}{s-1} + \frac{1}{2} \psi\left(\frac{s}{2}\right) \right] - \frac{1}{2} \log \pi, \quad A = -\operatorname{Re} \frac{\zeta'}{\zeta}(s), \quad (2)$$

the ledger of [44], with archimedean side G and prime side A . The region theorem of [44] states that $J > 0$ if $|x| \leq 3 \cdot 10^{12} - \frac{1}{2}$ or $y \geq \frac{1}{2} - 1/(5.573412 \log(|x| + \frac{1}{2}))$.

Comparison functions. Let $q_\epsilon(m, n) = 2m^2 + 3n^2$, $r_\epsilon(n) = \#\{(m, k) \in \mathbb{Z}^2 : q_\epsilon(m, k) = n\}$, $Z_\epsilon(s) = \sum_{n \geq 1} r_\epsilon(n) n^{-s}$ and $\xi_\epsilon(s) = s(s-1)(\sqrt{6}/\pi)^s \Gamma(s) Z_\epsilon(s)$. By [44, Lemma 16.1], ξ_ϵ is entire, $\xi_\epsilon(s) = \xi_\epsilon(1-s)$, and $\Xi_\epsilon(z) = \xi_\epsilon(\frac{1}{2} + iz) = \int_0^\infty \Phi_\epsilon(u) \cos(zu) du$ with the even positive kernel $\Phi_\epsilon(u) = 2 \sum' a(a-2)e^{u/2-a}$, $a = \pi q_\epsilon(m, n)e^u/\sqrt{6}$; genus theory gives $Z_\epsilon = \zeta L(\cdot, \chi_{-24}) - L(\cdot, \chi_{-3})L(\cdot, \chi_8)$. It has the zero $s_0 = 1.15958745297694888 + 9.45225081186590043i$ [44]. Let $\tan \vartheta = (\sqrt{10} - 2\sqrt{5} - 2)/(\sqrt{5} - 1)$ and $f(s) = \sum a(n) n^{-s}$ with $a = (1, \tan \vartheta, -\tan \vartheta, -1, 0)$ repeated with period 5, the Davenport–Heilbronn function [12]; $F_{\text{DH}}(s) = (5/\pi)^{(s+1)/2} \Gamma(\frac{s+1}{2}) f(s)$ satisfies $F_{\text{DH}}(s) = F_{\text{DH}}(1-s)$, and $\Xi_{\text{DH}}(z) = F_{\text{DH}}(\frac{1}{2} + iz)$. Both Ξ_ϵ and Ξ_{DH} are even entire functions of order one, real on $\mathbb{R}$; their zeros lie in $|\operatorname{Im} z| < 1.6$ and $|\operatorname{Im} z| < 2$ respectively (Propositions 10.1 and 10.3), so Proposition 2.1 applies to them. The zero $0.808517 + 85.699348i$ of f is due to Spira [53].

3 The flow picture

The function $\phi = \log |\Xi|$ is subharmonic, with $\Delta \phi = 2\pi \sum_w m_w \delta_w$, where m_w is the multiplicity. Its gradient $(\partial_x \phi, \partial_y \phi) = (u, v)$ is an irrotational, incompressible flow with a point source of strength $2\pi m_w$ at every zero, and $\Xi'/\Xi = u - iv$. We call $v > 0$ an updraft and $v < 0$ a downdraft; $J = 4Hv$ has the sign of v .

Theorem 3.1 (no isolated downdraft). *Let $F \not\equiv 0$ be entire and real on $\mathbb{R}$, and $v_F = \partial_y \log |F|$.*

- (i) *Near a real zero c of multiplicity k , $v_F(z) = ky/|z - c|^2 + O(y)$, so $v_F > 0$ on a half-disk around c minus c .*
- (ii) *Every bounded connected component Ω of $\{z \in \mathbb{C}_+ : F(z) \neq 0, v_F(z) < 0\}$ has a zero of F in $\mathbb{C}_+$ on its boundary.*
- (iii) *Every zero w of F in $\mathbb{C}_+$ lies on the boundary of $\{v_F < 0\}$.*

Proof. (i) Write $F(z) = (z - c)^k F_1(z)$ with $F_1(c) \neq 0$. Then $v_F = k \partial_y \log |z - c| + \partial_y \log |F_1| = ky/|z - c|^2 + v_{F_1}$, and v_{F_1} is harmonic near c , real-analytic, and vanishes on $\mathbb{R}$ because F_1 is real there; hence $|v_{F_1}| \leq Cy$. (ii) Since F is real on $\mathbb{R}$, $v_F(x, 0) = -\operatorname{Im}(F'/F)(x) = 0$ at every real x with $F(x) \neq 0$, and v_F is harmonic in a neighbourhood of such points. Suppose that $\bar{\Omega}$ contains no zero of F in $\mathbb{C}_+$. By (i) it contains no real zero either. Then v_F is harmonic in Ω , continuous on $\bar{\Omega}$, and $v_F = 0$ on $\partial\Omega$: a boundary point in $\mathbb{C}_+$ has $v_F = 0$ by maximality of Ω , and a boundary point on $\mathbb{R}$ has $v_F = 0$ by the above. Since Ω is bounded, the maximum principle gives $v_F \equiv 0$ on Ω , a contradiction. (iii) Near $w = c + ib$ of multiplicity k , $v_F = k(y - b)/|z - w|^2 + O(1)$, which tends to $-\infty$ along $x = c$, $y \uparrow b$. $\square$

Theorem 3.1 is Theorem 1.3(i). The boundary values do not control the interior: for the Davenport–Heilbronn function on the box $[85.2, 86.2] \times (0, 1]$, $v > 2.37$ on the top, $v > 0.008$ on the two sides and $v = 0$ on the bottom, while $v < 0$ on a region inside the box, below Spira’s zero $85.699348 + 0.308517i$ (Fig. 1c).

3.1 The Jensen–Riesz formula in a strip

For $Y > 0$ let $\mathbb{S}_Y = \{0 < y < Y\}$. For $z_0 = x_0 + iy_0 \in \mathbb{S}_Y$ and $t \in \mathbb{R}$ put $\alpha = \pi y_0/Y$, $\tau = \pi(x_0 - t)/Y$,

$$P_0(z_0, t) = \frac{\sin \alpha}{2Y(\cosh \tau - \cos \alpha)}, \quad P_Y(z_0, t) = \frac{\sin \alpha}{2Y(\cosh \tau + \cos \alpha)},$$

$$G_Y(z, w) = \log |e^{\pi(z - \bar{w})/Y} - 1| - \log |e^{\pi(z - w)/Y} - 1|.$$

The map $\zeta = e^{\pi z/Y}$ sends $\mathbb{S}_Y$ onto $\mathbb{C}_+$, the bottom edge onto $(0, \infty)$ and the top edge onto $(-\infty, 0)$. Pulling back the Poisson kernel $\pi^{-1} \operatorname{Im} \zeta_0/|\zeta_0 - \zeta|^2$ and the Green function $\log |(\zeta - \bar{\omega})/(\zeta - \omega)|$ of $\mathbb{C}_+$ gives P_0 , P_Y and G_Y ; in particular $G_Y > 0$ on $\mathbb{S}_Y \times \mathbb{S}_Y$ and $G_Y(z, w) = 0$ for $z \in \partial\mathbb{S}_Y$.

Theorem 3.2 (Jensen–Riesz formula). *Let $Y > \frac{1}{2}$ and $\phi = \log |\Xi|$. For every $z_0 \in \mathbb{S}_Y$ that is not a zero,*

$$\phi(z_0) = \int_{\mathbb{R}} P_0(z_0, t) \phi(t) dt + \int_{\mathbb{R}} P_Y(z_0, t) \phi(t + iY) dt - \sum_{w \in \mathbb{S}_Y} m_w G_Y(z_0, w).$$

Consequently the defect $D(z_0) = \mathcal{P}_Y[\phi](z_0) - \phi(z_0)$, where $\mathcal{P}_Y[\phi](z_0)$ is the sum of the two integrals, is nonnegative, and RH holds if and only if $D(z_0) = 0$ at a single point $z_0 \in \mathbb{S}_Y$.

Proof. The zeros of Ξ in $\mathbb{S}_Y$ are those in $\mathbb{C}_+$, since $|\operatorname{Im} w| < \frac{1}{2} < Y$, and there are $O(\log X)$ of them with $|\operatorname{Re} w| \in [X, X + 1]$. For fixed z , $G_Y(z, w) \leq Ce^{-\pi|\operatorname{Re}(z - w)|/Y}$, so $\Pi(z) = \sum_w m_w G_Y(z, w)$ converges locally uniformly on $\mathbb{S}_Y$ minus the zeros and is $O(\log(2 + |x|))$ at distance ≥ 1 from them. The function $\psi = \phi + \Pi$ is harmonic in $\mathbb{S}_Y$: each $m_w G_Y(\cdot, w)$ has the logarithmic singularity $-m_w \log |z - w|$ plus a harmonic function, cancelling the singularity of ϕ at w [40, Ch. 4]. On $\partial\mathbb{S}_Y$, $\Pi = 0$ and $\psi = \phi$; the boundary values are continuous except for logarithmic singularities at the real zeros, $\phi(t + iY) = -\frac{\pi}{4}|t| + O(\log |t|)$ because $\operatorname{Re} s = \frac{1}{2} + Y > 1$ there, and $\int_X^{X+1} |\phi(t)| dt = O(X)$ by Jensen’s formula on disks of radius 2. Hence the Poisson integral $\mathcal{P} = \mathcal{P}_Y[\phi]$ converges, since $P_0, P_Y = O(e^{-\pi|t|/Y})$, and $\mathcal{P} - \psi$ is harmonic in $\mathbb{S}_Y$, tends to 0 at every boundary point (at a real zero c both $\mathcal{P}$ and ψ differ from $m_c \log |z - c|$ by functions continuous at c), and is $O(|x|)$. A harmonic function on a strip of width Y with zero boundary values and growth $o(e^{\pi|x|/Y})$ vanishes [40]. Hence $\psi = \mathcal{P}$, which is the formula. Since $G_Y > 0$, $D(z_0) = \sum m_w G_Y(z_0, w)$ vanishes if and only if Ξ has no zero in $\mathbb{S}_Y$, that is, no zero in $\mathbb{C}_+$, which is RH. $\square$

Theorem 3.2 is Theorem 1.2(a). Balazard, Saias and Yor [1] proved the half-plane form $\frac{1}{2\pi} \int_{\mathbb{R}} \log |\zeta(\frac{1}{2} + it)| \frac{dt}{1+t^2} = \sum_{\beta > 1/2} \log |\frac{\rho}{1-\rho}|$; Theorem 3.2 localises the defect to a single point of a strip and gives it as a sum of Green functions over the off-line zeros. Differentiating in y_0 gives $v(z_0) = \partial_{y_0} \mathcal{P}(z_0) - \sum_w m_w \partial_{y_0} G_Y(z_0, w)$. For the comparison functions, evaluated by quadrature of the Poisson integrals with $Y = 1$, the defect equals the Green sum to 1.1×10^{-10} for Ξ_{DH} at $85.7 + 0.15i$ ($D = 0.9845$) and to 3×10^{-14} for Ξ_{ϵ} at $9.45 + 0.3i$ ($D = 0.6229$); for Ξ , $|D| \leq 3 \times 10^{-6}$ at $14.13 + 0.25i$, $7005.08 + 0.3i$ and $17143.8 + 0.2i$, the size of the truncation error of the Poisson integrals.

3.2 Measure of the downdraft

Theorem 3.3. *Theorem 1.3(ii) holds.*

Proof. By Proposition 2.1, $\{J \leq 0\} \cap \mathbb{C}_+$ lies in the union of the closed half-disks $|z - \gamma| \leq b$ over the zeros $\gamma + ib$ of Ξ with $b > 0$. A half-disk meeting $[T, 2T] \times (0, \infty)$ has $\gamma \in [T - \frac{1}{2}, 2T + \frac{1}{2}]$, and it has area $\frac{\pi}{2} b^2$ and horizontal section of length $2\sqrt{b^2 - y^2} \leq 2b \mathbf{1}_{b > y}$ at height y . Let $n(\eta)$ be the number of such zeros with $b > \eta$; these correspond to zeros ρ of ζ with $\beta > \frac{1}{2} + \eta$ and $\gamma \in [T - \frac{1}{2}, 2T + \frac{1}{2}]$, so Selberg's density theorem $N(\sigma, T) \ll T^{1-(\sigma-1/2)/4} \log T$ [43] gives $n(\eta) \ll T^{1-\eta/4} \log T$. Then

$$\sum b^2 = \int_0^{1/2} 2\eta n(\eta) d\eta \ll T \log T \int_0^\infty 2\eta e^{-\frac{\eta}{4} \log T} d\eta = \frac{32 T}{\log T},$$

which gives the first bound. The half-disks with $b > \eta$ have total area at most $\frac{\pi}{8} n(\eta)$, which gives the second. For the third, the section at height y has measure at most $\sum_{b > y} 2b = 2 \int_y^{1/2} n(\eta) d\eta + 2y n(y) \ll T^{1-y/4} (8 + 2y \log T)$. $\square$

With $y \geq A \log \log T / \log T$ the third bound is $\ll T(\log T)^{-A/4} (A \log \log T + 4)$.

Proposition 3.4. *Theorem 1.3(iii) holds.*

Proof. Ford proved $\zeta(\beta + it) \neq 0$ for $1 - \beta \leq 1/(57.54(\log |t|)^{2/3}(\log \log |t|)^{1/3})$, $|t| \geq 3$ [16, Thm. 5]. Let $F(t) = (\log t)^{2/3}(\log \log t)^{1/3}$, increasing for $t \geq 3$. If $J(x, y) \leq 0$ with $y > 0$, Proposition 2.1 gives a zero $\gamma + ib$ of Ξ with $b > 0$ and $(x - \gamma)^2 + y^2 \leq b^2$. Then $|\gamma| > 3 \cdot 10^{12}$, $|\gamma| < |x| + \frac{1}{2}$, and $b = \beta' - \frac{1}{2}$ for the zero $\beta' + i\gamma$ of ζ with $\beta' > \frac{1}{2}$, so $y \leq b < \frac{1}{2} - 1/(57.54F(|\gamma|)) < \frac{1}{2} - 1/(57.54F(|x| + \frac{1}{2}))$. The width $1/(57.54F(t))$ exceeds $1/(5.573412 \log t)$ exactly when $\log t / \log \log t > (57.54/5.573412)^3$, that is, for $\log t > 10151.5$. $\square$

4 The surgery obstruction

We prove Theorem 1.1 and illustrate it at the two Lehmer pairs below height 2×10^4 .

Proof of Theorem 1.1. (i) $(z - \gamma_1)(z - \gamma_2) = (z - \gamma)^2 - a^2$, so $R(z) = 1 + (a^2 + b^2)/((z - \gamma)^2 - a^2)$ is rational with real coefficients and $R(\infty) = 1$, and $R(z)R(-z)$ is even. The poles of $R(z)R(-z)$ are $\pm \gamma_1, \pm \gamma_2$, which are zeros of Ξ , so $\tilde{\Xi}$ is entire; it is even and real on $\mathbb{R}$, and of order one because Ξ is. Its zeros are as stated. The new zeros $\pm \gamma \pm ib$ have $|\operatorname{Re}| > 3 \cdot 10^{12}$ and $b \leq \frac{1}{4} < \frac{1}{2} - 1/(5.573412 \log \gamma)$. The counting function changes only by the removal of γ_1 and γ_2 and the addition of $\gamma \pm ib$, both at real part $\gamma \in (\gamma_1, \gamma_2)$. (ii) By (i), the zeros of $\tilde{\Xi}$ satisfy the two hypotheses from which the region theorem of [44] is proved: the half-disks of Proposition 2.1 have centres $|\gamma'| > 3 \cdot 10^{12}$ and radii below $\frac{1}{2} - 1/(5.573412 \log |\gamma'|)$. So $\tilde{J} > 0$ off $\mathcal{U}$. On the line $x = \gamma$,

$\partial_y \log |\tilde{\Xi}| = \partial_y \log |\Xi| + \partial_y \log |R(z)R(-z)|$, and $\partial_y \log |R(\gamma + iy)| = \partial_y [\log |b^2 - y^2| - \log(y^2 + a^2)] = -2y/(b^2 - y^2) - 2y/(y^2 + a^2) \rightarrow -\infty$ as $y \uparrow b$, while $\partial_y \log |\Xi(\gamma + iy)|$ and $\partial_y \log |R(-\gamma - iy)|$ are bounded above there (a zero of Ξ near $\gamma + ib$ contributes $(y - \operatorname{Im} w)/|z - w|^2$, which is bounded above as $y \uparrow b$ unless $w = \gamma + ib$, when it tends to $-\infty$). (iii) Put $w = z - \gamma$, so $\operatorname{Im} w = y$. Since $|w - a| \geq y$ and $|w + a| \geq y$, $|w^2 - a^2| \geq y^2$, hence $|R(z) - 1| \leq (a^2 + b^2)/y^2 = \varepsilon$ and $|\log |R(z)|| \leq -\log(1 - \varepsilon) \leq \varepsilon/(1 - \varepsilon)$. The same holds for $R(-z)$ with $w = -z - \gamma$, which gives the first bound. For the second, $(\log R)' = 2w/(w^2 + b^2) - 2w/(w^2 - a^2) = -2w(a^2 + b^2)/((w^2 + b^2)(w^2 - a^2))$; for $|w| \geq y \geq 2a \geq 2b$ both factors of the denominator exceed $\frac{3}{4}|w|^2$ in modulus, so $|(\log R)'| \leq \frac{32}{9}(a^2 + b^2)|w|^{-3} \leq \frac{32}{9}(a^2 + b^2)y^{-3}$, and likewise for $R(-z)$; finally $|\nabla \log |F|| = |F'/F|$. (iv) Selberg's theorem gives $N_0(T + U) - N_0(T) \gg U \log T$ for the number N_0 of zeros of odd order on the critical line, for $U = T$ and T large [55, Ch. X]. So $[T, 2T]$ contains $\gg T \log T$ distinct real zeros of Ξ , and two of them lie within $2C/\log T$ of each other. Let $y \geq \frac{1}{2}$ (which is $\operatorname{Re} s \geq 1$) and T be large, so that $2a \leq \frac{1}{2}$. If $|x| \leq 4T$, then $\varepsilon \leq 8a^2$ and the two bounds of (iii) are at most $32a^2$ and $\frac{64}{9} \cdot 16a^2$, both $\ll 1/\log^2 T \ll 1/\log^2(|x| + T)$. If $|x| > 4T$, then $w = z - \gamma$ and $w = -z - \gamma$ satisfy $|w| \geq |x|/2 > 2T$, so $|w^2 - a^2|$ and $|w^2 + b^2|$ exceed $\frac{1}{2}|w|^2$; hence $|R - 1| \leq 16a^2/|x|^2$ and $|(\log R)'| \leq 128a^2/|x|^3$ at z and at $-z$, which are $\ll 1/\log^2(|x| + T)$. For the last assertion, suppose that inequalities for $\log |\Xi|$ and $\nabla \log |\Xi|$ on $\operatorname{Re} s \geq 1$ hold with a margin $C_1/\log^2(|x| + 2)$. By (iv), with $T \geq 2$, they hold, without the margin, for $\tilde{\Xi}$, and by (i) $\tilde{\Xi}$ has the same verified zeros, zero-free region, counting function up to ± 1 and growth; but $\tilde{J} < 0$ at points of $\mathcal{U}$ by (ii). $\square$

Remark 4.1 (the surgered function has no Euler product). Put $r(s) = R(z)R(-z)$ with $z = -i(s - \frac{1}{2})$ and $\tilde{\zeta} = \zeta r$, so that $\tilde{\Xi}(z) = \frac{1}{2}s(s - 1)\pi^{-s/2}\Gamma(\frac{s}{2})\tilde{\zeta}(s)$. If $\tilde{\zeta}$ were a Dirichlet series absolutely convergent in some half-plane, so would be $r = \tilde{\zeta} \cdot (1/\zeta)$. A Dirichlet series is almost periodic in t on vertical lines, while $r(\sigma + it) \rightarrow 1$ as $|t| \rightarrow \infty$ and $r \not\equiv 1$; hence $\tilde{\zeta}$ is not a Dirichlet series, and in particular has no Euler product. A property of Ξ that fails for every $\tilde{\Xi}$ therefore uses information about ζ beyond the hypotheses of Theorem 1.1, such as its Dirichlet series and Euler product.

The Lehmer pairs. The consecutive zeros $\gamma_1 = 7005.062866174917$, $\gamma_2 = 7005.100564672642$ have $a = 0.0188492$, and 17143.786536184, 17143.821843505 have $a = 0.0176537$. These lie below $3 \cdot 10^{12}$, and the surgery there illustrates (iii) at the sizes of (iv). With $b = a$, $\max_x |\log |\tilde{\Xi}/\Xi||$ is 2.842×10^{-3} on $y = \frac{1}{2}$ ($\operatorname{Re} s = 1$) and 7.11×10^{-4} on $y = 1$, against the bound $2\varepsilon/(1 - \varepsilon)$ with $\varepsilon = 2.84 \times 10^{-3}$ and 7.1×10^{-4} ; $\max |\partial_y \log |R(z)R(-z)||$ is 1.14×10^{-2} and 1.4×10^{-3} , while $\min v \geq 2.229$ on $y = \frac{1}{2}$, $|x - \gamma| \leq 3$. Outside the new half-disk $\partial_y \log |\tilde{\Xi}| > 0$, as Proposition 2.1 requires, and at $(\gamma, 0.99b)$ it equals -5278.5 . At the pair near 17143.8 the values are 2.49×10^{-3} and -5636 ; with $b = a/4$ at 7005 they are 1.51×10^{-3} and -21114 (Fig. 1).

5 Operator forms

An entire function E is of Hermite–Biehler class if $|E(\bar{z})| < |E(z)|$ for $z \in \mathbb{C}_+$; then $K_E(w, z) = (E(z)\overline{E(w)} - E^\#(z)\overline{E^\#(w)})/(2\pi i(\bar{w} - z))$, $E^\#(z) = \overline{E(\bar{z})}$, is the reproducing kernel of a de Branges space [13].

Proposition 5.1 (the Hermite–Biehler threshold). *Theorem 1.5(i) holds.*

Proof. By Hadamard's product for the even function Ξ of order one (Proposition 2.1), which makes no assumption on the location of the zeros,

$$\frac{|E_h(z)|^2}{|E_h(\bar{z})|^2} = \prod_w \frac{|x - c_w + i(h + y - \delta_w)|^2}{|x - c_w + i(h - y - \delta_w)|^2}, \quad w = c_w + i\delta_w,$$

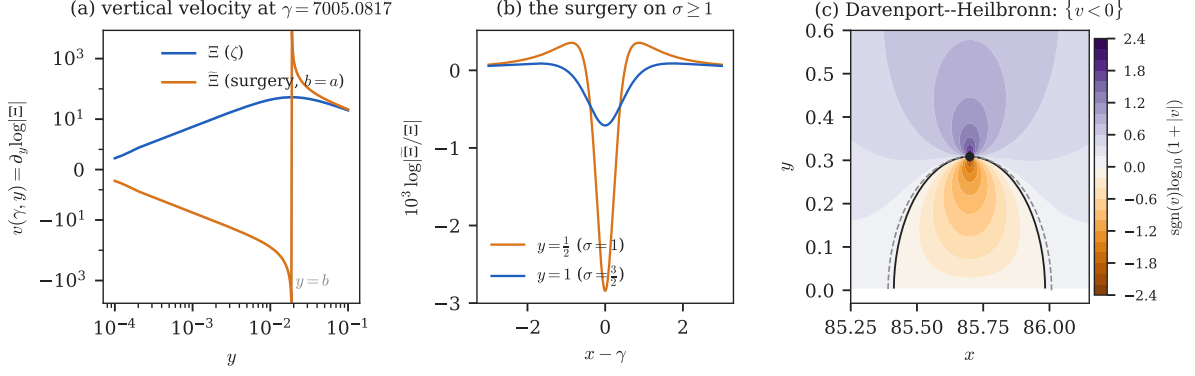


Figure 1: (a) The vertical velocity $v = \partial_y \log |\Xi|$ on the line through the midpoint $\gamma = 7005.0817$ of the Lehmer pair, for Ξ and for the surgered function $\tilde{\Xi}$ with $b = a = 0.01885$; v is positive for Ξ and falls to $-\infty$ below the planted zero at $y = b$ for $\tilde{\Xi}$. (b) The change $\log |\tilde{\Xi}/\Xi|$ on $\text{Re } s = 1$ and $\text{Re } s = \frac{3}{2}$: at most 2.84×10^{-3} . (c) The downdraft of the Davenport–Heilbronn function below Spira’s zero $85.6993 + 0.3085i$ (dot): colour gives $\text{sgn}(v) \log_{10}(1 + |v|)$, the solid curve is $v = 0$, and the dashed curve is the half-disk of radius 0.3085 of Proposition 2.1.

with conjugate zeros grouped so that the product converges. A factor exceeds 1 exactly when $(h + y - \delta_w)^2 > (h - y - \delta_w)^2$, that is when $4y(h - \delta_w) > 0$. If $h \geq \Theta = \sup \delta_w$, every factor is at least 1 for $y > 0$, and the factors with $\delta_w < 0$, which exist, exceed 1; so E_h is of Hermite–Biehler class. If $h < \Theta$, some zero has $\delta_w > h$, and E_h vanishes at $w - ih \in \mathbb{C}_+$, where $|E_h(\bar{z})| < |E_h(z)|$ fails. $\Theta \leq \frac{1}{2}$ because $\zeta(s) \neq 0$ for $\text{Re } s \geq 1$ [55, Ch. III]. $\square$

The threshold is due in substance to Lagarias [26], whose theorem that the zeros of $\xi(s + h) \pm \xi(s - h)$ lie on the critical line for $|h| \geq \frac{1}{2}$ rests on this Hermite–Biehler property, read in de Branges’s theory.

Proposition 5.2 (the current as a Hermite–Biehler defect). *Theorem 1.5(ii) holds.*

Proof. $E_J^\# = \Xi - i\Xi'$ because Ξ is real on $\mathbb{R}$. Expanding,

$$E_J(z)\overline{E_J(w)} - E_J^\#(z)\overline{E_J^\#(w)} = 2i[\Xi'(z)\overline{\Xi(w)} - \Xi(z)\overline{\Xi'(w)}],$$

so $K_{E_J}(w, z) = (\Xi'(z)\overline{\Xi(w)} - \Xi(z)\overline{\Xi'(w)})/(\pi(\bar{w} - z))$. With $m = -\Xi'/\Xi$, $\Xi(z)N_m(z, w)\overline{\Xi(w)} = (-\Xi'(z)\overline{\Xi(w)} + \Xi(z)\overline{\Xi'(w)})/(z - \bar{w})$, which is $\pi K_{E_J}(w, z)$. At $w = z$, $|E_J(z)|^2 - |E_J(\bar{z})|^2 = 2i \cdot 2i \text{Im}(\Xi'\Xi) = 4 \text{Im}(\Xi\Xi') = -4 \text{Im}(\Xi\Xi'') = J$, since $J = 2\partial_y |\Xi|^2 = 4 \text{Re}(\Xi i\Xi')$. On $\mathbb{R}$ the difference quotient gives $\pi K(x, x) = \Xi'^2 - \Xi\Xi''$. $\square$

Given $J > 0$ off $\mathcal{U}$ [44], Conjecture 1 holds if and only if E_J is of Hermite–Biehler class. The number $\pi K_{E_J}(x, x) = \Xi'^2 - \Xi\Xi''$ is the Laguerre expression, $\frac{1}{16} \int Q(p, x)p^2 dp$ in the phase-space form of [44].

Theorem 5.3 (the index of the current). *The kernels N_m on $\mathbb{C}_+$ minus the zeros of Ξ , and K_{E_J} on $\mathbb{C}_+$, have exactly N_+ negative squares.*

Proof. Lower bound. Let $a_1, \dots, a_k$ be distinct zeros of Ξ in $\mathbb{C}_+$ of multiplicities μ_j . Near a_j , $m(z) = -\mu_j/(z - a_j) + O(1)$. For small $\varepsilon > 0$ put $z_j = a_j - i\mu_j\varepsilon \in \mathbb{C}_+$, so that $m(z_j) = -i/\varepsilon + O(1)$. Then

$$N_m(z_i, z_j) = \frac{-2i/\varepsilon + O(1)}{a_i - \bar{a}_j + O(\varepsilon)} = -\frac{2}{\varepsilon} \frac{i}{a_i - \bar{a}_j} + O(1).$$

The matrix $C_{ij} = i/(a_i - \bar{a}_j) = \int_0^\infty e^{ia_it} \overline{e^{ia_jt}} dt$ is the Gram matrix of the linearly independent functions e^{ia_jt} in $L^2(0, \infty)$, hence positive definite. So $[N_m(z_i, z_j)]$ has k negative eigenvalues for small ε , and the number of negative squares is at least $\min(k, N_+)$ for every k .

Upper bound. Let $N_+ < \infty$ and let P be the real even polynomial $\prod(1 - z/w)$ over the non-real zeros w with multiplicity. Then $\Xi_0 = \Xi/P$ is even, real on $\mathbb{R}$, of order one with only real zeros, so $\Xi_0 = \Xi_0(0) \prod_{c>0} (1 - z^2/c^2)$ and $m_0 = -\Xi'_0/\Xi_0 = \sum_{c>0} [(c-z)^{-1} + (-c-z)^{-1}]$; each term $(c-z)^{-1}$, $c \in \mathbb{R}$, has the kernel $(c-z)^{-1} \overline{(c-w)^{-1}} \succeq 0$, so $N_{m_0} \succeq 0$. The remainder $-P'/P$ is $\sum_a h_a$ over the distinct zeros $a \in \mathbb{C}_+$ with $h_a(z) = \mu_a[(a-z)^{-1} + (\bar{a}-z)^{-1}]$, and a direct computation gives

$$N_{h_a}(z, w) = \mu_a [f_a(z) \overline{f_{\bar{a}}(w)} + f_{\bar{a}}(z) \overline{f_a(w)}], \quad f_a(z) = (a-z)^{-1},$$

whose quadratic form $\mu_a \cdot 2 \operatorname{Re}(A\bar{B}) = \frac{\mu_a}{2}(|A+B|^2 - |A-B|^2)$ has at most one negative square. Since the number of negative eigenvalues of a sum of Hermitian matrices is at most the sum of the numbers, $N_m = N_{m_0} + \sum_a N_{h_a}$ has at most N_+ negative squares.

From N_m to K_{E_J} . By Proposition 5.2, on points where $\Xi \neq 0$ the matrix of πK_{E_J} is DND^* with $D = \operatorname{diag}(\Xi(z_i))$ invertible, which has the same inertia. K_{E_J} is continuous on $\mathbb{C}_+ \times \mathbb{C}_+$ and the zeros of Ξ are isolated, so moving the points z_i that are zeros slightly keeps every negative eigenvalue of $[K_{E_J}(z_i, z_j)]$ negative; hence the count on $\mathbb{C}_+$ equals the count on $\mathbb{C}_+$ minus the zeros. $\square$

Theorem 5.3 is Theorem 1.5(iii) and Theorem 1.2(b), since RH is $N_+ = 0$. Krein and Langer [24] proved that a function of the generalised Nevanlinna class N_κ has at most κ poles in $\mathbb{C}_+$; the theorem identifies the index of the current with the number N_+ of distinct zeros of Ξ in $\mathbb{C}_+$. Bombieri [4] proved the analogous count for Weil's form: if finitely many zeros fail RH, the number of negative eigenvalues of its truncations, for large truncations, is half the number of these zeros.

Remark 5.4 (a self-adjoint operator for the zeros). If $N_+ < \infty$ and Ξ has no multiple real zero, E_J has no real zeros and K_{E_J} is the reproducing kernel of a Pontryagin space of entire functions of negative index N_+ [21], which for $N_+ = 0$ is the de Branges space $\mathcal{H}(E_J)$ [13]. Multiplication by z in it is symmetric with deficiency indices $(1, 1)$, and its self-adjoint extension associated with $A = \frac{1}{2}(E_J + E_J^\#) = \Xi$ has the zeros of Ξ as spectrum. A self-adjoint operator with the zeros of Ξ as spectrum thus exists in a space with an indefinite metric of index N_+ , and the metric is definite exactly under RH.

Corollary 5.5. *Theorem 1.5(iv) holds.*

Proof. Where $\Xi \neq 0$, $\Xi + it\Xi' = 0$ if and only if $m(z) = -i/t$, and then $v = \operatorname{Im} m = -1/t < 0$, so $J < 0$; $\{J < 0\} \subset \mathcal{U}$ by [44]. By Proposition 2.1, $v > 0$ in $\mathbb{C}_+$ outside the finitely many half-disks over the zeros off the line, and $v \rightarrow 0$ at the real axis away from real zeros; so for t in a compact subset of $(0, \infty)$ the solutions of $m = -i/t$ lie in a fixed compact subset of $\mathbb{C}_+$ that contains no zero of Ξ on its boundary. The zeros of $m + i/t = i(\Xi + it\Xi')/(t\Xi)$ in $\mathbb{C}_+$ therefore move continuously with t and their number is constant on $(0, \infty)$. As $t \rightarrow 0$, near each distinct zero a of multiplicity μ there is exactly one solution, $z = a - i\mu t + O(t^2)$ (Rouché's theorem for $-\mu/(z-a) + O(1) = -i/t$), and no other solution remains in the compact set. So the number is N_+ . $\square$

Theorem 5.6 (shadow law). *Theorem 1.5(v) holds: under its hypotheses, with $\mu_0 = u'(c)/u(c) \in \mathbb{R}$, $\Xi + it\Xi'$ has exactly one zero in $|z - c| < Cb^2$, at $c - \frac{1}{2}\mu_0 b^2 + ib^2/2t + O(b^4)$, where C and the implied constant depend only on t and on bounds for u'/u and its derivative on the disk.*

Proof. With $\omega = z - c$, $m(z) = -2\omega/(\omega^2 + b^2) - u'/u(z)$, and $u'/u(c + \omega) = \mu_0 + O(\omega)$. For $|\omega| \leq Cb^2$, $\omega^2 + b^2 = b^2(1 + O(b^2))$, and $m = -i/t$ becomes $\omega = -\frac{b^2}{2}(\mu_0 - i/t) + O(b^4)$. Rouché's theorem on $|\omega| = Cb^2$ with $C > |\mu_0 - i/t|$ gives exactly one solution. $\square$

At the surgery of Section 4 at 7005.0817 with $t = 1$, the zero of $\tilde{\Xi} + i\tilde{\Xi}'$ lies at $7005.081715 + 2.000 \times 10^{-6}i$ for $b = 0.002$, and its height divided by $b^2/2$ is 1.0000, 1.0000, 1.0004, 1.0097, 1.0895 for $b = 0.002, 0.005, 0.02, 0.1, 0.3$. The matrices $[N_m(z_i, z_j)]$ on 15 points near the planted zero have exactly one negative eigenvalue, from -3.6×10^{-2} at $b = 0.002$ to -38.5 at $b = 0.3$, and on 12 points near $\pm\gamma$ exactly two (-1.2909 twice at $b = 0.02$), while for Ξ itself the smallest eigenvalue is $+9.2 \times 10^{-10}$, at the level of the conditioning. For Ξ_ϵ and Ξ_{DH} the zeros of $F + iF'$ near the first off-line zeros are $9.64697122 + 0.29436266i$ and $85.74318239 + 0.05152712i$.

5.1 De Branges's positivity condition

De Branges [14] proposed, for $E(z) = \xi(1 - iz)$, the condition $\text{Re}\langle F(z), F(z + i) \rangle_{\mathcal{H}(E)} \geq 0$ for all F with $F, F(\cdot + i) \in \mathcal{H}(E)$, which implies RH. Conrey and Li [9] derived from it the necessary condition $-\text{Re}\{\xi'(\rho)\xi(1 + \rho)\} \geq 0$ at every zero ρ and found the value -5.389×10^{-69} at the 34-th zero in the normalisation $s(s - 1)\pi^{-s/2}\Gamma(\frac{s}{2})\zeta(s)$ of ξ .

Theorem 5.7. *Theorem 1.9(i) holds.*

Proof. Write $\xi = g\zeta$ and $Z(t) = e^{i\theta(t)}\zeta(\frac{1}{2} + it)$. At $s = \frac{1}{2} + it$, $\frac{1}{2}s(s - 1) = -\frac{1}{2}(t^2 + \frac{1}{4})$ and $\pi^{-s/2}\Gamma(\frac{s}{2}) = \pi^{-1/4}|\Gamma(\frac{1}{4} + \frac{it}{2})|e^{i\theta(t)}$, so $g(\frac{1}{2} + it) = -|g|e^{i\theta(t)}$. Differentiating $\zeta(\frac{1}{2} + it) = e^{-i\theta}Z$ at a zero gives $\zeta'(\rho) = -ie^{-i\theta(\gamma)}Z'(\gamma)$, hence $\xi'(\rho) = g(\rho)\zeta'(\rho) = i|g(\rho)|Z'(\gamma)$ and $-\text{Re}\{\xi'(\rho)\xi(1 + \rho)\} = |g(\rho)|Z'(\gamma)\text{Im}\xi(\frac{3}{2} + i\gamma)$. Moreover $\Xi'(\gamma) = i\xi'(\rho) = -|g|Z'(\gamma)$ and $\Xi(\gamma + i) = \xi(-\frac{1}{2} + i\gamma) = \xi(\frac{3}{2} - i\gamma) = \overline{\xi(\frac{3}{2} + i\gamma)}$, which gives the second form. For the sign: if the zeros up to γ_n are simple and on the line, $\text{sgn} Z'(\gamma_n) = (-1)^{n+1}$ because $Z(0) = \zeta(\frac{1}{2}) < 0$. By Stirling's formula $\arg \Gamma(\sigma + i\tau) = \tau \log \tau - \tau + (\sigma - \frac{1}{2})\frac{\pi}{2} + O(1/\tau)$, so $\arg g(\frac{3}{2} + it) = \theta(t) + \frac{5\pi}{4} + O(1/t)$ modulo 2π . With $\theta(\gamma_n) = \pi(n - \frac{3}{2} - S(\gamma_n))$ (the Riemann-von Mangoldt formula at the midpoint of the jump) and $a_n = \arg \zeta(\frac{3}{2} + i\gamma_n)$, $\arg \xi(\frac{3}{2} + i\gamma_n) = \pi n - \frac{\pi}{4} - \pi S(\gamma_n) + a_n + O(1/\gamma_n)$, so $\text{sgn} \text{Im} \xi(\frac{3}{2} + i\gamma_n) = (-1)^{n+1} \text{sgn} \sin(\frac{\pi}{4} + \pi S(\gamma_n) - a_n + O(1/\gamma_n))$, and the product with $\text{sgn} Z'(\gamma_n)$ is the stated sign. $\square$

Over the first 2469 zeros, up to height 2999.5, the condition fails at 252 (10.21%), first at $n = 34$, $\rho = \frac{1}{2} + 111.0295355431697i$, where $-\text{Re}\{\xi'(\rho)\xi(1 + \rho)\} = -1.347 \times 10^{-69} = -5.389 \times 10^{-69}/4$, the value of Conrey and Li in our normalisation. The failure fraction in consecutive blocks of about 500 zeros is 0.064, 0.096, 0.124, 0.106, 0.122, and the sign of $\sin(\frac{\pi}{4} + \pi S(\gamma_n) - a_n)$ agrees with that of the condition at all but two of the 2469 zeros. The standard deviation of $S(\gamma_n)$ over these zeros is 0.272. We also reproduce $\text{Re}\{\xi(1 + 282i)/\xi(2 + 282i)\} = -1.31957 \times 10^{-4}$ [9].

5.2 The Husimi current and the archimedean side

Theorem 5.8. *Theorem 1.9(ii) holds.*

Proof. Let $W(a, k) = \frac{1}{2\pi} \int K(a + \frac{q}{2})K(a - \frac{q}{2})e^{-ikq} dq$ be the Wigner function of K . As in [44, Prop. 11.1], $H_F(k, y) = \frac{\pi}{2} \int W(a, k) \cosh(2ya) da$ and $J_F(k, y) = 2\pi \int W(a, k) a \sinh(2ya) da$. Let G be the centred Gaussian density on $\mathbb{R}^2$ with variances $\sigma_a^2 = s/2$ in a and $\sigma_k^2 = 1/2s$ in k , so that $\sigma_a\sigma_k = \frac{1}{2}$. Then $W * G$ is the Husimi function of K , $(W * G)(a, k) = c|\langle \phi_{a,k}, K \rangle|^2$ with $\phi_{a,k}$ the Gaussian wave packet of widths (σ_a, σ_k) centred at (a, k) and $c > 0$ [7]; it is

real-analytic, nonnegative and vanishes only on a discrete set when $K \neq 0$. For the nonnegative symbol $\sigma(a, k) = a \sinh(2ya) \delta(k - x)$,

$$\begin{aligned} 0 &< \iint (W * G) \sigma = \iint W (\sigma * G) \\ &= \sqrt{\frac{s}{\pi}} e^{y^2 s} \int e^{-s(k-x)^2} \iint W(a, k) [a \sinh(2ya) + ys \cosh(2ya)] da dk, \end{aligned}$$

because $\mathbb{E}[(a + g) \sinh(2y(a + g))] = e^{2y^2 \sigma_a^2} (a \sinh(2ya) + 2y \sigma_a^2 \cosh(2ya))$ for $g \sim N(0, \sigma_a^2)$. The inner integral is $(J_F + 4ysH_F)(k, y)/2\pi$. $\square$

The inequality holds for every real even kernel, Φ_ϵ included. At $x_0 = 9.45225$, $y = 0.3$, $J_\epsilon = -1.031 \times 10^{-8}$ and $H_\epsilon = 1.593 \times 10^{-9}$, so the pointwise quantity $J_\epsilon + 4ysH_\epsilon$ is negative for $s < 5.39$, while the Gaussian average of Theorem 1.9(ii) is positive for every s between 0.05 and 500 that we evaluated. The observable $\hat{O}_{x,y}$ of [44, 46] with Weyl symbol $a \sinh(2ya) \delta(k - x) \geq 0$ is not a positive operator, since its expectation in the state $\Phi_\epsilon / \|\Phi_\epsilon\|$ is proportional to J_ϵ , which is negative at $(x_0, 0.3)$; its anti-Wick quantisation, the Gaussian average of Theorem 5.8, is positive for every state.

Proposition 5.9. *Theorem 1.9(iii) holds. Moreover each Macdonald function $K_{ix}(2\pi N e^p)$ of the expansion of Q in [44] satisfies $-f'' + (2\pi N)^2 e^{2p} f = x^2 f$, and for fixed $\alpha > 0$ the function $x \mapsto K_{ix}(\alpha)$ has only real zeros.*

Proof. At $s = \frac{1}{2} + ix$, $\frac{1}{s} + \frac{1}{s-1} = 2ix/(-x^2 - \frac{1}{4})$ is purely imaginary, so $G(x, 0) = \frac{1}{2} \operatorname{Re} \psi(\frac{1}{4} + \frac{ix}{2}) - \frac{1}{2} \log \pi$, and $\theta'(x) = \frac{d}{dx} \operatorname{Im} \log \Gamma(\frac{1}{4} + \frac{ix}{2}) - \frac{1}{2} \log \pi$ is the same. The differential equation is Bessel's equation $z^2 K'' + zK' - (z^2 - x^2)K = 0$ in the variable $p = \log(z/2\pi N)$. The reality of the zeros is Pólya's theorem [38]. $\square$

By the Riemann–von Mangoldt formula $N(t) = \theta(t)/\pi + 1 + S(t)$, $\theta(t)/\pi + 1$ is the smooth part of the counting function of the zeros, which Berry and Keating [2] obtained from the regularised Hamiltonian xp , and θ'/π is its density; off the zeros $v(x, 0) = 0$, so on the line the prime side also equals $\theta'(x)$, as $-\operatorname{Re} \zeta'/\zeta(\frac{1}{2} + ix) = \theta'(x)$ shows directly. At the first zero, $G(\gamma_1, 0) = \theta'(\gamma_1) = 0.4052743722$.

6 The heat flow

Let $\Phi_T(u) = e^{Tu^2/4} \Phi(u)$ and $\Xi_T(z) = \int_0^\infty \Phi_T(u) \cos(zu) du$. In the normalisation of Polymath [39], $\Xi_T(z) = 8H_T(2z)$ with $H_t(z) = \int_0^\infty e^{tu^2} \Phi_P(u) \cos(zu) du$ [46], so Λ is the least T with Ξ_T real-rooted for all larger T , $0 \leq \Lambda \leq 0.2$ [41, 37], and de Bruijn's theorem [15, 39] reads: if the zeros of Ξ_{T_0} satisfy $|\operatorname{Im} z| \leq \Delta$, those of Ξ_T , $T \geq T_0$, satisfy $|\operatorname{Im} z|^2 \leq \max(\Delta^2 - (T - T_0)/2, 0)$. Differentiating under the integral, $\partial_T \Xi_T = -\frac{1}{4} \Xi_T''$, and a simple zero moves with velocity $\dot{z} = \frac{1}{4} \Xi_T'' / \Xi_T'(z) = \frac{1}{2} \sum_{z' \neq z} (z - z')^{-1}$. Put $\mathcal{L}_T = \Xi_T'^2 - \Xi_T \Xi_T''$.

Theorem 6.1. (i) *If $T > \Lambda$, then $\mathcal{L}_T > 0$ on $\mathbb{R}$.*

(ii) *If $\Lambda > 0$, then Ξ_Λ has a real zero of multiplicity at least 2.*

Hence Theorem 1.2(c) holds.

Proof. (i) For $T > \Lambda$ the zeros of Ξ_T are real and simple [11, Cor. 1]. Ξ_T is even, and of order one in z , hence of order at most $\frac{1}{2}$ in z^2 , so $\Xi_T = \Xi_T(0) \prod_j (1 - z^2/x_j^2)$ with $\Xi_T(0) > 0$. Off the zeros, $\mathcal{L}_T = \Xi_T'^2 \cdot (- (\Xi_T'/\Xi_T)') = \Xi_T'^2 \sum_j [(x - x_j)^{-2} + (x + x_j)^{-2}] > 0$, and at a simple zero $\mathcal{L}_T = \Xi_T'^2 > 0$. (ii) Take $T_n \uparrow \Lambda$, $T_n > 0$, and non-real zeros z_n of Ξ_{T_n} . By de Bruijn's theorem

$|\operatorname{Im} z_n| \leq \frac{1}{2}$, and by Theorem 1.5(i) of [39], which makes the result of Ki, Kim and Lee [23] uniform, the non-real zeros of Ξ_T have $|\operatorname{Re} z| \leq X(T_0)$ for $0 < T_0 \leq T \leq \frac{1}{2}$; we take $T_0 = \Lambda/2$ and $T_n \geq T_0$. A subsequence converges to some z^* . Since $\Xi_{T_n} \rightarrow \Xi_\Lambda$ locally uniformly and Ξ_Λ is real-rooted, z^* is real. By Hurwitz's theorem the two distinct zeros $z_n, \bar{z}_n$ converge to z^* , which is therefore a zero of multiplicity at least 2. For Theorem 1.2(c): RH gives $\Lambda = 0$ by [41], and then (i) and [11, Cor. 1] give $\mathcal{L}_T > 0$ and simple zeros for all $T > 0$. If RH fails, Ξ_0 has a non-real zero, so $\Lambda > 0$, $\Lambda \leq 0.2$ [37], and by (ii) Ξ_Λ has a multiple real zero x^* , where $\mathcal{L}_\Lambda(x^*) = 0$. The formula for Λ follows from (i), (ii) and [11, Cor. 1]. $\square$

On $[0, 40]$ the relative Laguerre margin $\min_x \mathcal{L}_T / (\Xi_T'^2 + |\Xi_T \Xi_T''|)$ is 0.1741, 0.1737, 0.1732, 0.1708, 0.1685, 0.1641 at $T = 0.2, 0.1, 0, -0.5, -1, -2$.

Theorem 6.2 (Hermite local law). *Theorem 1.6(i) holds.*

Proof. Let $\tau = (T - T^*)/4$. For entire functions of order one, $\Xi_T = \sum_{k \geq 0} (-\tau)^k \Xi_{T^*}^{(2k)} / k!$, the series converging locally uniformly (Cauchy's estimates with $\log \max_{|z|=r} |\Xi_{T^*}| = O(r \log r)$) give $|\Xi_{T^*}^{(2k)}| \leq (Ck^{-1} \log k)^{2k} (2k)!$ on compact sets). Near x^* , $\Xi_{T^*}(x^* + \omega) = c \omega^m (1 + O(\omega))$ with $c \neq 0$ real. Since $e^{-t^2/2} e^{\omega t} = \sum \operatorname{He}_j(\omega) t^j / j!$, the operator $\sum_k (-\tau)^k \partial^{2k} / k!$ sends ω^m to $(2\tau)^{m/2} \operatorname{He}_m(\omega / \sqrt{2\tau})$, and ω^{m+j} to a polynomial of size $O(|\tau|^{(m+j)/2})$ for $\omega = O(|\tau|^{1/2})$. Hence $\Xi_T(x^* + \sqrt{2|\tau|} \eta) = c(2|\tau|)^{m/2} [\operatorname{He}_m^\pm(\eta) + O(|\tau|^{1/2})]$ uniformly for η in compact sets, with $\operatorname{He}_m^+ = \operatorname{He}_m$ for $\tau > 0$ and $\operatorname{He}_m^-(\eta) = i^m \operatorname{He}_m(-i\eta)$ for $\tau < 0$. The zeros of He_m are real and simple, and those of He_m^- are i times them. Rouché's theorem places exactly one zero of Ξ_T near each, and by Hurwitz's theorem Ξ_T has exactly m zeros in a fixed small disk around x^* for T near T^* , so these are all of them. For $\tau > 0$ each is simple and alone in a disk symmetric about $\mathbb{R}$; since Ξ_T is real on $\mathbb{R}$, it is real. For $\tau < 0$ only the zero near $\eta = 0$ (present when m is odd) can be real. Finally $2|\tau| = |T - T^*|/2$. $\square$

Theorem 6.3 (landing). *Theorem 1.6(ii) holds.*

Proof. The zeros are $x_T \pm ib_T$ with $b_T^2 = (T^* - T)/2(1 + o(1))$ by Theorem 6.2 with $\operatorname{He}_2(\eta) = \eta^2 - 1$. Write $\Xi_T = qu$ near x^* with $q = (z - x_T)^2 + b_T^2$ and u holomorphic, zero-free and real on $\mathbb{R}$, uniformly for T near T^* . For real x , $\mathcal{L}[qu] = u^2 \mathcal{L}[q] + q^2 \mathcal{L}[u]$ and $\mathcal{L}[q] = 2(x - x_T)^2 - 2b_T^2$, while $q^2 = O(b_T^4)$ where $|x - x_T| \leq 2b_T$; so $\mathcal{L}_T < 0$ exactly on an interval of half-width $b_T(1 + O(b_T^2))$. For $z = x + iy$, $d = x - x_T$, $\partial_y \log |q| = 2y(d^2 + y^2 - b_T^2)/|q|^2$ and $\partial_y \log |u| = \kappa_0 y + O(y^3)$ with κ_0 bounded. Where $d^2 + y^2 \leq 4b_T^2$, $|q|^2 \leq 81b_T^4$, so v has the sign of $d^2 + y^2 - b_T^2 + O(b_T^4)$. Where $4b_T^2 \leq d^2 + y^2 \leq \delta_0^2$, with δ_0 small and fixed, $d^2 + y^2 - b_T^2 \geq \frac{3}{4}(d^2 + y^2)$ while $|q|^2 \leq 2(d^2 + y^2)^2$, so $v > 0$ there. Hence near x^* the boundary of $\{J_T < 0\}$ is the circle $|z - x_T| = b_T$ displaced by $O(b_T^3)$. $\square$

For Ξ_ϵ the first off-line pair lands at $T_c = 0.747230705896$ at $x^* = 8.993177745521$. At $T = 0.745$ the set $\{\mathcal{L}_T < 0\}$ has half-width 0.03338 against $b_T = 0.03339$ and $\sqrt{(T_c - T)/2} = 0.03340$; for $T_c < T \leq T_c + 10^{-3}$ the two real zeros satisfy $\delta^2 / (2(T - T_c)) = 1.0000$ (10^{-5} , 10^{-4}) and 0.9998 (10^{-3}), with δ their separation. Backwards, the Lehmer pairs collide at $T_c = -7.112047 \times 10^{-4}$ (gap $\delta_0 = 0.037698$ at 7005.08) and $T_c = -6.241606 \times 10^{-4}$ ($\delta_0 = 0.035307$ at 17143.8), close to $-\delta_0^2/2$ as the two-zero model of Theorem 1.6(ii) gives.

Theorem 6.4 (two copies). *Theorem 1.6(iii) holds.*

Proof. $H(x, y) = \frac{1}{4} \iint \Phi(u) \Phi(v) e^{izu - i\bar{z}v} du dv$ with $izu - i\bar{z}v = ix(u - v) - y(u + v)$ [44]. Write $\Phi = e^{-Tu^2/4} \Phi_T$ and $-T(u^2 + v^2)/4 = -T(u - v)^2/4 - Tuv/2$. Since $e^{-T(u-v)^2/4} e^{ix(u-v)} = \mathbb{E} e^{i(x+g)(u-v)}$ for $g \sim N(0, T/2)$, $H(x, y) = \mathbb{E} \mathcal{Z}_T(x + g, y)$. Expanding $e^{-Tuv/2} = \sum_k (-T/2)^k u^k v^k / k!$, which is dominated because Φ_T decays doubly exponentially, and using

$\frac{1}{2} \int \Phi_T(u)(iu)^k e^{izu} du = \Xi_T^{(k)}(z)$ gives the series. Differentiating in y gives the statement for J . $\square$

At $T = 0.2$ the identity holds to 1.6×10^{-15} , 6.9×10^{-12} and 2.6×10^{-9} relative at $(5, 0.05)$, $(14.1347, 0.1)$ and $(21, 0.3)$. In the language of [44], the forward flow is a positive merge and the backward flow is not:

Proposition 6.5. *For $T > 0$, $e^{Tu^2/4} = \mathbb{E} e^{ug}$ with $g \sim N(0, T/2)$, while $e^{-Tu^2/4}$ is not of the form $\int e^{ug} \mu(dg)$ for any positive measure μ .*

Proof. The first is the Gaussian moment generating function. By Hölder's inequality, $u \mapsto \log \int e^{ug} \mu(dg)$ is convex, while $-Tu^2/4$ is strictly concave. $\square$

Theorem 6.6 (three-state chains). *Theorem 1.6(iv) holds.*

Proof. Let $S = \sum_i S_i$, $S_i \in \{-1, 0, 1\}$, with weight $e^K \sum S_i S_{i+1}$. Since $e^{TS^2/4} = \mathbb{E} e^{\sqrt{T/2} Sg}$ and $TS^2/4 = \frac{T}{4} \sum_i S_i^2 + \frac{T}{2} \sum_{i < j} S_i S_j$, $Z_{n,T}$ is the partition function of three-state spins with single-site weights $(e^{T/4}, 1, e^{T/4})$ and couplings $K \geq 0$ (neighbours) and $T/2 \geq 0$ (all pairs). A spin with weights $(w, 1, w)$ is the merged pair of two spins $\pm \frac{1}{2}$ with coupling $e^{4\kappa_0 \varsigma_1 \varsigma_2}$, $w = e^{2\kappa_0}/2$ [44]; here $\kappa_0 = \frac{1}{2} \log 2 + T/8 \geq 0$. So $Z_{n,T}$ is the partition function of a spin- $\frac{1}{2}$ ferromagnet, and its zeros lie on $|z| = 1$ by the Lee–Yang theorem [27, 17]. For $n = 1$, $Z_{1,T} = e^{T/4}(z + z^{-1}) + 1$ vanishes where $z + z^{-1} = -e^{-T/4}$, which lies in $[-2, 2]$ exactly when $T \geq -4 \log 2$. For $K = 0$ and $n \geq 2$, $Z_{n,0} = (z + 1 + z^{-1})^n$ has zeros of multiplicity n at $e^{\pm 2\pi i/3}$. On $|z| = 1$, $z = e^{i\vartheta}$, $\partial_T Z_{n,T} = -\frac{1}{4} \partial_\vartheta^2 Z_{n,T}$, so Theorem 6.2 applies in ϑ : for $T < 0$ close to 0 at most one of the n zeros near $e^{2\pi i/3}$ is on the circle, and $\Lambda_n(0) = 0$. For $n = 2$, with $c = z + z^{-1}$, $Z_{2,T} = e^{T+K}(c^2 - 2) + 2e^{T/4}c + 1 + 2e^{-K}$, a quadratic in c whose discriminant is $4q \Delta(q)$ with $q = e^{T/2}$ and

$$\Delta(q) = 2e^{2K}q^3 - (e^K + 2)q + 1 = (q - e^{-K})(2e^{2K}q^2 + 2e^Kq - e^K).$$

For $K < \log 4$ the positive root of the second factor is below e^{-K} , so $\Delta > 0$ for $T > -2K$ and $\Delta < 0$ for T slightly below $-2K$. For $T \geq -2K$ the vertex $-e^{-3T/4-K}$ lies in $[-2, 0)$ and the value at $c = -2$, $2e^{T+K} - 4e^{T/4} + 1 + 2e^{-K}$, is increasing in T there and equals $(2e^{-K/2} - 1)^2 \geq 0$ at $T = -2K$, while the value at $c = 2$ is positive; so both roots c lie in $[-2, 2]$, which puts all four zeros on $|z| = 1$. Hence $\Lambda_2(K) = -2K$. Finally, if $K > 0$ and the $2n$ zeros of $Z_{n,0}$ on the circle are simple, they stay simple and, by the symmetry $z \mapsto 1/\bar{z}$ of the zero set, on the circle for T in a neighbourhood of 0; so $\Lambda_n(K) < 0$. $\square$

K	$n = 2$	$n = 4$	$n = 8$	$n = 12$	$n = 16$
0.1	-0.2	-0.05272	-0.006991	-0.001854	-0.0006582
0.5	-1.0	-0.2445	-0.04452	-0.01231	-0.004685
1	-2.0	-0.5063	-0.09434	-0.02965	-0.01258

Table 2: $\Lambda_n(K)$ for chains of n three-state spins, computed by bisection in T with the zeros on $|z| = 1$ counted by Sturm sequences. The column $n = 2$ is $-2K$ exactly.

7 Weil's functional and Li's coefficients

For a function h holomorphic in a neighbourhood of the zeros, $W(h) = \sum_\rho h(\gamma_\rho)$ with $\rho = \frac{1}{2} + i\gamma_\rho$ (γ_ρ is complex when ρ is off the line), summed with ρ and $\bar{\rho}$ grouped.

Theorem 7.1 (the Weil–Poisson form). *Let $y > 0$, $s = \frac{1}{2} + y + ix$, $\xi(s) \neq 0$, $h = h_{x,y}$ and $\varphi = \varphi_{x,y}$.*

- (i) $J = 4HW(h)$, $h = |\widehat{\varphi}|^2$ with $\widehat{\varphi}(r) = \int \varphi(u)e^{iru} du = \sqrt{y}/(y + i(x - r))$, and $h = \widehat{f}$ for $f = \varphi * \tilde{\varphi} = \frac{1}{2}e^{-y|u|}e^{-ixu}$, $\tilde{\varphi}(u) = \varphi(-u)$.
- (ii) For $y > \frac{1}{2}$ Weil’s explicit formula holds for f with an absolutely convergent prime sum and is the ledger (2): the pole terms give $h(\frac{i}{2}) + h(-\frac{i}{2}) = \text{Re}[\frac{1}{s} + \frac{1}{s-1}]$, the archimedean term gives $\frac{1}{2} \text{Re} \psi(\frac{s}{2}) - \frac{1}{2} \log \pi$, and the prime term gives $A = \sum \Lambda(n)n^{-1/2-y} \cos(x \log n)$.
- (iii) For $0 < y \leq \frac{1}{2}$, $\sum \Lambda(n)n^{-1/2}|f(\log n)| = \infty$ and h has poles at $x \pm iy$ in $|\text{Im } r| \leq \frac{1}{2}$. The functions $\sum_{n \leq X} \Lambda(n)n^{-s} - X^{1-s}/(1-s)$ converge to $-\zeta'/\zeta(s)$, whose real part is A , locally uniformly on $\{\text{Re } s > \frac{1}{2} + y_0\}$ if and only if $\Theta \leq y_0$.

Proof. (i) For a real zero γ , $h(\gamma)$ is the term $y/((x - \gamma)^2 + y^2)$ of Proposition 2.1. For a pair $c \pm ib$, $h(r) = \frac{1}{2i}[(\bar{z} - r)^{-1} - (z - r)^{-1}]$ continues analytically, and $h(c + ib) + h(c - ib) = -\text{Im}(z - c - ib)^{-1} - \text{Im}(z - c + ib)^{-1}$, which is the pair term. The formulas for $\widehat{\varphi}$ and f are direct integrations. (ii) Is the logarithmic derivative of the Hadamard product combined with $\xi'/\xi = \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2} \log \pi + \frac{1}{2} \psi(\frac{s}{2}) + \zeta'/\zeta$, whose Dirichlet series converges absolutely for $\text{Re } s > 1$; the pole terms are computed from $h(r) = \frac{1}{2}[(y + i(x - r))^{-1} + (y - i(x - r))^{-1}]$ at $r = \pm \frac{i}{2}$. (iii) The divergence is $\sum \Lambda(n)n^{-1/2-y} = \infty$ for $y \leq \frac{1}{2}$. Partial summation with $\psi(t) = t + E(t)$ gives $\sum_{n \leq X} \Lambda(n)n^{-s} - X^{1-s}/(1-s) = -s/(1-s) + E(X)X^{-s} + s \int_1^X E(t)t^{-s-1} dt$, and $E(t) = O(t^{\frac{1}{2} + \Theta + \epsilon})$ gives convergence to $-\zeta'/\zeta(s)$ for $\text{Re } s > \frac{1}{2} + \Theta$, locally uniformly. Conversely, a locally uniform limit on $\{\text{Re } s > \frac{1}{2} + y_0\}$ is holomorphic there and equals $-\zeta'/\zeta$ on $\text{Re } s > 1$, so ζ has no zero with $\beta > \frac{1}{2} + y_0$. $\square$

Theorem 7.1(i) with [44] gives Theorem 1.2(d): $\int Q p \sinh(y p) dp = 16HW(h_{x,y})$, so Conjecture 1 is Weil’s positivity for the Poisson family $\varphi_{x,y}$, $(x, y) \in \mathcal{U}$. Indeed, under RH Ξ has no zero in $\mathbb{C}_+$ and $W(h_{x,y}) = J/4H > 0$ there; conversely, if Ξ has no zero in $\mathcal{U}$ and $W(h_{x,y}) > 0$ on $\mathcal{U}$, then $H > 0$ and $J = 4HW(h_{x,y}) > 0$ on $\mathcal{U}$, which is Conjecture 1. The zero side and $\text{Re } \xi'/\xi$ agree to 3.3×10^{-8} at twelve points with $0 \leq x \leq 1500$, $10^{-3} \leq y \leq 0.4$, computed from the 2469 zeros below 3000 with a density tail, and to 6.9×10^{-8} relative at the Lehmer pair near 7005; the explicit formula at (50, 1.5) agrees to 2.9×10^{-9} .

Theorem 7.2 (the Li–current identity). *With $w = 1 - 1/s$,*

$$J = 4H \text{Re} \left[\frac{1}{s(s-1)} \sum_{n \geq 1} \lambda_n w^n \right],$$

the series converging absolutely for $|w| < R_\zeta = \min(1, \inf_{\beta > 1/2} |1 - 1/\rho|)$. RH holds if and only if $R_\zeta = 1$, and unconditionally $R_\zeta^2 \geq 1 - 1/T_0^2$ with $T_0 = 3 \cdot 10^{12}$.

Proof. With $F(w) = \log \xi(1/(1-w))$, $F'(w) = \sum_{n \geq 0} \lambda_{n+1} w^n$ [22, 5], and $F'(w) = s^2 \xi'/\xi(s)$ for $s = 1/(1-w)$, so $\xi'/\xi(s) = s^{-2} w^{-1} \sum \lambda_n w^n$ and $s^2 w = s(s-1)$. The singularities of F' in $|w| \leq 1$ are the points $1 - 1/\rho$ and $w = 1$ ($s = \infty$), and $|1 - 1/\rho| < 1$ exactly when $\beta > \frac{1}{2}$. For $\beta > \frac{1}{2}$, $|1 - 1/\rho|^2 = 1 - (2\beta - 1)/(\beta^2 + \gamma^2) > 1 - 1/\gamma^2$, and $|\gamma| > T_0$. At $s = \frac{1}{2} + y + ix$, $|w|^2 = 1 - 2y/((y + \frac{1}{2})^2 + x^2) < 1$ for $y > 0$. $\square$

Theorem 7.2 is Theorem 1.2(e). On $\mathcal{U}$ the series needs about $|s|^2/y \approx 10^{26}$ terms. It reproduces $\text{Re } \xi'/\xi$ to ten digits at (0, 0.3), (2, 0.2), (3, 0.5) and $(\gamma_1, 0.3)$ with the coefficients below.

Lemma 7.3 (insensitivity of the Li coefficients). *The estimate displayed in Theorem 1.4 holds.*

Proof. Let $\phi(\beta) = n \log(1 - 1/(\beta + i\gamma))$ and $\Psi(b) = e^{\phi(\frac{1}{2}+b)} + e^{\phi(\frac{1}{2}-b)}$, the quantity in question being $\Psi(b) - \Psi(0)$ since $1 - \bar{\rho} = \frac{1}{2} - b + i\gamma$. Ψ is even, so $|\Psi(b) - \Psi(0)| \leq \frac{b^2}{2} \max_{|b'| \leq b} |\Psi''(b')|$. Now $\phi'(\beta) = n/(\rho(\rho - 1))$ with $|\rho(\rho - 1)| \geq \gamma^2$, so $|\phi'| \leq n/\gamma^2$; $\phi'' = -n(2\rho - 1)/(\rho(\rho - 1))^2$, so $|\phi''| \leq n(1 + 2\gamma)/\gamma^4 \leq 3n/\gamma^3$; and $|1 - 1/\rho|^2 = 1 - (2\beta - 1)/(\beta^2 + \gamma^2) \leq 1 + 2b/\gamma^2$, so $|e^\phi| \leq e^{nb/\gamma^2} \leq e^{1/2}$ for $n \leq \gamma^2$. Hence $|\Psi''| \leq 2e^{1/2}(n^2\gamma^{-4} + 3n\gamma^{-3})$. $\square$

Proof of Theorem 1.4. $\lambda_n = \sum_{\rho} [1 - (1 - 1/\rho)^n]$, with ρ and $\bar{\rho}$ grouped. For $\rho_0 = \frac{1}{2} + i\gamma$, $1 - 1/\rho_0 = e^{i\theta_\gamma}$ with $\theta_\gamma = 2 \arctan(1/2\gamma)$, so a conjugate pair on the line contributes $2(1 - \cos n\theta_\gamma) \geq 0$. Let $\lambda_n^{(0)}$ be the value obtained when every zero above $T_0 = 3 \cdot 10^{12}$ is moved onto the line with its ordinate kept; every term of $\lambda_n^{(0)}$ is nonnegative. By Lemma 7.3 with $b < \frac{1}{2}$, and since each quadruple off the line has two zeros with positive ordinate,

$$|\lambda_n - \lambda_n^{(0)}| \leq \frac{e^{1/2}}{2} \sum_{\gamma > T_0} (n^2\gamma^{-4} + 3n\gamma^{-3}).$$

Trudgian's bound $|N(t) - \frac{t}{2\pi} \log \frac{t}{2\pi e} - \frac{7}{8}| \leq 0.112 \log t + 0.278 \log \log t + 2.51 + 0.2/t$ [56] gives $N(t) \leq \frac{t}{2\pi} \log t$ for $t \geq 20$, and partial summation gives $\sum_{\gamma > T} \gamma^{-k} \leq \frac{k}{2\pi(k-1)} (\log T + 1) T^{1-k}$. With $T_0 = 3 \cdot 10^{12}$ this bounds the error by $1.93 \times 10^{-37} n^2 + 1.96 \times 10^{-24} n$, which is 2.2×10^{-12} at $n = 10^{12}$. For the main term keep one pair. If $5 \leq n \leq 10^{12}$, there is an ordinate $\gamma \in [n, 3n] \subset (0, T_0]$: for $5 \leq n \leq 29$ one of 14.1347, 21.0220, 30.4249 lies there, and for $n \geq 30$ Trudgian's bound gives $N(3n) - N(n) \geq \bar{N}(3n) - \bar{N}(n) - E(3n) - E(n)$, with $\bar{N}(t) = \frac{t}{2\pi} \log \frac{t}{2\pi e} + \frac{7}{8}$ and $E(t)$ the right side of Trudgian's bound; this lower bound is 14.4 at $n = 30$ and increasing in n , since for $n \geq 30$ the derivative of $\bar{N}(3n) - \bar{N}(n)$ exceeds 1 while that of $E(3n) + E(n)$ is below 0.02. Then $n\theta_\gamma \in [2n \arctan \frac{1}{6n}, 2n \arctan \frac{1}{2n}] \subset [0.3332, 1]$, so $\lambda_n^{(0)} \geq 2(1 - \cos 0.3332) = 0.1099$. For $n \leq 4$ the first zero gives $n\theta_{\gamma_1} \in [0.0707, 0.283]$ and $\lambda_n^{(0)} \geq 2(1 - \cos 0.0707) = 0.0049$. $\square$

Computed coefficients. By a Cauchy integral of F' on $|w| = 0.9998$ with 2^{18} nodes, which uses no zeros, $\lambda_n > 0$ for $n \leq 30000$, with minimum $\lambda_1 = 0.0230957090$ (exactly $1 + \frac{\gamma_E}{2} - \frac{1}{2} \log 4\pi = 0.02309570896612$), $\lambda_2 = 0.0923457352$, $\lambda_3 = 0.2076389206$, $\lambda_{100} = 118.6038$, $\lambda_{1000} = 2326.0532$ and $\lambda_{30000} = 120740.3$, against Keiper's asymptotic $\frac{n}{2}(\log \frac{n}{2\pi} - 1 + \gamma_E) = 120724.4$ [22]. For $n \leq 1200$ a second integral on $|w| = 0.99$ with 8192 nodes agrees to 2.3×10^{-8} , and the sum over the zeros below 3000 with a density tail agrees within its error bound.

Windows. Weil's functional restricted to test functions of bounded support was studied by Bombieri [4]. For a finite-dimensional space V of real functions supported in $[-L/2, L/2]$ let

$$\mu(V) = \min_{0 \neq g \in V} \frac{W(\widehat{g * \tilde{g}})}{\|g\|_2^2}, \quad W(\widehat{g * \tilde{g}}) = \frac{1}{\pi} \int_{\mathbb{R}} \widehat{g}(t - i\delta) \overline{\widehat{g}(t + i\delta)} \frac{\xi'}{\xi}(c + it) dt, \quad \delta = c - \frac{1}{2},$$

the contour integral holding for every $c > \frac{1}{2} + \Theta$ by the argument principle and the functional equation. A negative $\mu(V)$ shows that Weil's form is not positive on windows of length L . We use the spaces $V_h(L)$ of cubic B-splines with knot spacing at most h , and $V_h(L, T_0)$ spanned by $B_j(u) \cos(T_0 u)$ and $B_j(u) \sin(T_0 u)$. For ζ on $V_{0.1}(L)$, with $c = 1.5$ and $c = 2.6$ agreeing and agreeing with the zero sum, $\mu = 0.312, 9.57 \times 10^{-3}, 2.32 \times 10^{-5}, 1.51 \times 10^{-9}, 3.13 \times 10^{-11}, 2.68 \times 10^{-12}, 7.9 \times 10^{-13}$ at $L = 0.3, 0.69, 1, 2, 3, 4, 5$. Connes and Consani proved positivity at the archimedean place for g supported in $[2^{-1/2}, 2^{1/2}]$ multiplicatively [8], that is, $L = \log 2$ here.

8 The merge

Theorem 8.1 (truncations). *Theorem 1.7(i) holds.*

Proof. Let ϕ_n be the n -th term of (1) and $\mathcal{T} = \Phi - \Phi^b$ the omitted part, a sum of ϕ_n with $n \geq 2$. Then $\phi'_n(0) = 4e^{-\pi n^2}(-4\pi^3 n^6 + 15\pi^2 n^4 - 7.5\pi n^2) < 0$ for $n \geq 2$, so $\mathcal{T}'(0) < 0$. For the Epstein kernel each term is $2a(a-2)e^{u/2-a}$, $a = c_0 Q e^u$, $c_0 = \pi/\sqrt{6}$, with derivative $2a(-a^2 + 4.5a - 3)e^{-a}$ at $u = 0$, negative for $a > 3.69$; the omitted forms have $Q \geq 3$, $a \geq 3c_0 = 3.85$. Integrating by parts twice, $\int_0^\infty \mathcal{T}(u) \cos(xu) du = -\mathcal{T}'(0)/x^2 + o(x^{-2})$, while $\Xi(x)$ and $\Xi_\epsilon(x)$ decay exponentially. So $F(x) = \mathcal{T}'(0)x^{-2}(1 + o(1)) < 0$ for large x , and F , which is analytic, has finitely many real zeros. Each term of Φ^b is positive for $u \geq 0$, and restricting to $u \in [U, U+1]$ with $2\pi e^{2U} = y$ in $F(iy) = \int_0^\infty \Phi^b \cosh(yu) du$ gives $\log F(iy) \geq \frac{y}{2} \log y - O(y)$. So F is entire of order one and infinite type, hence has infinitely many zeros by Hadamard's theorem, all but finitely many of them non-real. $\square$

The real zeros of the truncations at $N = 1, 2, 3, 4$ number 1, 7, 15, 31, the last at 14.044, 39.532, 65.03, 103.37, and each truncation has non-real zeros with real part below 130 (3, 4, 6, 7 in $\text{Im } z < 12$). The heights are close to $4(N+1)^2$, where $e^{-\pi x/4}$ and $e^{-\pi(N+1)^2}$ balance. The 3-smooth and 5-smooth truncations have the same 25 and 43 real zeros as Ξ below 90 and 130, since their omitted tails start at $n = 5$ and $n = 7$. The Epstein truncations with $V = 20$ and 30 reproduce the real zeros 5.131, 12.075, 15.063 of Ξ_ϵ and its off-line zero near $9.452 + 0.660i$.

Proposition 8.2 (evenness). *Theorem 1.7(ii) holds. Consequently a kernel whose cosine transform has infinitely many real zeros has all odd derivatives zero at 0.*

Proof. Repeated integration by parts gives $\int_0^\infty f \cos(xu) du = \sum_{j < k_0} (-1)^{j+1} f^{(2j+1)}(0) x^{-2j-2} + O(x^{-2k_0-2})$ for every k_0 . If k is the least index with $f^{(2k+1)}(0) \neq 0$, the transform is $(-1)^{k+1} f^{(2k+1)}(0) x^{-2k-2} (1 + O(x^{-2})) \neq 0$ for large $|x|$. It is analytic in $|\text{Im } z| < c$, so it has finitely many zeros in every compact interval. $\square$

Theorem 8.3 (no cosh factor). *Theorem 1.7(iii) holds. Evaluated with 4000 terms of (1) on a grid of step 5×10^{-5} , $\Phi(iy)$ is real, positive on $[0, y^*)$, has a single zero $y^* = 0.638830$ in $(0, 0.7739]$, where it changes sign, and is negative on $(y^*, 0.7739]$, its smallest absolute value on the grid there, 1.6×10^{-7} , being at the right end; so the hypothesis holds for every $a \geq 2.03$ except $a^* = \pi/2y^* = 2.45886$.*

Proof. Φ is analytic and doubly-exponentially decaying in $|\text{Im } u| < \pi/4$. For $a > 2$ the first pole of $\text{sech}(au)$ above $\mathbb{R}$ is $i\pi/2a$, inside this strip, with residue $-i/a$. Shifting $\int_{\mathbb{R}} \Phi(u) \text{sech}(au) e^{ixu} du$ to $\text{Im } u = c_a \in (\pi/2a, \min(3\pi/2a, \pi/4))$ gives $2\pi i \cdot (-i/a) \Phi(i\pi/2a) e^{-\pi x/2a} + O(e^{-c_a x})$, and half of it is the cosine integral. If $\Phi(i\pi/2a) \neq 0$, the transform F_a of the even, positive, doubly-exponentially decaying kernel $\Phi \text{sech}(au)$ has no real zeros for large $|x|$ and, as in Theorem 8.1, infinitely many zeros; so it is not in the Laguerre–Pólya class. If $\Phi = \cosh(au)\Phi_0$, then $\Phi_0 = \Phi \text{sech}(au)$. $\square$

Multiplying a kernel by $\cosh(au)$ preserves the reality of the zeros of its transform, since $\int \cosh(au) \Phi_0(u) \cos(zu) du = \frac{1}{2} [F_0(z+ia) + F_0(z-ia)]$ [15]; Theorem 8.3 shows that Φ has no such factor with $a > 2$, $a \neq a^*$. At $x = 30$ the ratio of F_a to its leading term is 1.0112, 1.0005, 1.00004 for $a = 3, 4, 6$. For $a = 0.5, 1, 1.5, 2, 2.5$ the transform F_a has 6, 6, 3, 1, 0 real zeros in $(0, 40)$ and 0, 0, 1, 2, 3 zeros in $(0, 40) \times (0, 4)$, against 6 and 0 for Ξ .

Theorem 8.4 (the ν -flow). *Theorem 1.7(iv) holds.*

Proof. For $\operatorname{Re} s < 0$, $\mathbb{E}[X^{s/2}] = \Gamma(-\frac{s}{2})^{-1} \int_0^\infty \beta^{-s/2-1} \mathbb{E} e^{-\beta X} d\beta$ with $\mathbb{E} e^{-\beta S_\nu} = (x/\sinh x)^\nu$, $x = \sqrt{2\beta}$ [44]. Expanding $(x/\sinh x)^\nu = (2x)^\nu \sum_k c_k(\nu) e^{-(\nu+2k)x}$, $c_k(\nu) = \Gamma(k+\nu)/(\Gamma(\nu)k!)$, and integrating term by term,

$$M_\nu(s) = \left(\frac{\pi}{2}\right)^{s/2} \frac{2^{1+s/2+\nu} \Gamma(\nu-s)}{\Gamma(-s/2)} D_\nu(\nu-s), \quad D_\nu(w) = \sum_{k \geq 0} c_k(\nu) (\nu+2k)^{-w},$$

first for $\operatorname{Re} s < 0$ and then by continuation. At $\nu = 2$, $c_k = k+1$ and $D_2(w) = 2^{-w} \zeta(w-1)$, and the duplication formula recovers $M_2 = 2\zeta$. Since $\partial_\nu c_k|_{\nu=2} = (k+1)(H_{k+1}-1)$ and $\partial_\nu (\nu+2k)^{-w} = -w(\nu+2k)^{-w-1}$,

$$\partial_\nu D_\nu(w)|_{\nu=2} = 2^{-w} [E(w-1) - \zeta(w-1) - \frac{w}{2} \zeta(w)], \quad E(s) = \sum_n H_n n^{-s} = -\zeta'(s) + \gamma_E \zeta(s) + \Upsilon(s).$$

At a zero ρ of ξ , $D_2(2-\rho) = 0$ because $\zeta(1-\rho) = 0$. Writing $M_\nu(s) = A_\nu(s) D_\nu(\nu-s)$ and differentiating, $d\rho/d\nu = -\partial_\nu M/\partial_s M = 1 + \partial_\nu D(2-\rho)/D'_2(2-\rho)$, with $D'_2(2-\rho) = 2^{\rho-2} \zeta'(1-\rho) \neq 0$ for a simple zero. Substituting gives the formula. The implicit function theorem applies because $\partial_s M_2(\rho) = 2\zeta'(\rho) \neq 0$. $\square$

The formula agrees with finite differences of $\log M_\nu$ in ν to eight digits at $s = 0.3 + 5i$, $-0.4 + 3i$ and $0.7 + 9i$. For the first ten zeros $\operatorname{Re} d\rho/d\nu = -0.567, +0.162, -0.982, +1.527, -2.162, +0.710, +0.862, -2.753, +4.174, -3.281$; at γ_1 it matches the slope $(0.443 - 0.556)/0.2 = -0.565$ of the zero path of [44]. Of the 1517 zeros with $10 < \gamma < 2000$, 761 (50.2%) move to the right as ν increases through 2. The members of each Lehmer pair move in opposite directions: +6078 and -6644 at 7005.06, 7005.10, and +23371 and -23088 at 17143.79, 17143.82. So near $\nu = 2$ the zeros of M_ν lie on both sides of the line, and the line is reached from both sides.

Theorem 8.5 (Hausdorff moments). *Theorem 1.2(f) holds.*

Proof. $\log \Xi(z) = \log \Xi(0) - \sum_{m \geq 1} S_m z^{2m}/m$ for $|z| < |w_1|$, from the product over the zeros with $\operatorname{Re} w > 0$; zeros come in pairs $w, \bar{w}$, so S_m is real. With $Z(h) = \Xi(-ih) = \frac{1}{2} \int \Phi(u) e^{hu} du$, the partition function of Riemann's spin [31, 44], the cumulants are $\kappa_{2m} = (-1)^{m+1} (2m)! S_m/m$. If RH holds, $S_{m+1} = \sum_\gamma \gamma^{-2} (\gamma^{-2})^m = \int_0^1 a^m \nu(da)$ with the finite positive measure $\nu = \sum_{\gamma > 0} \gamma^{-2} \delta_{\gamma^{-2}}$ on $[0, \gamma_1^{-2}]$. Conversely, if $S_{m+1} = \int_0^1 a^m \nu(da)$, then $\sum_m S_{m+1} x^m = \int \nu(da)/(1-xa)$ is holomorphic off $[1, \infty)$; but it equals $\sum_{\operatorname{Re} w > 0} (w^2 - x)^{-1}$ near 0, a meromorphic function with a pole at each distinct w^2 . So every w^2 lies in $[1, \infty)$ and every zero is real. Unconditionally, the zeros with $|\operatorname{Re} w| \leq 3 \cdot 10^{12}$ are real, and the others contribute at most $2 \sum_{|w| > 3 \cdot 10^{12}} |w|^{-2m} < \gamma_1^{-2m}$ in absolute value, so $S_m \geq \gamma_1^{-2m} - 2 \sum_{|w| > 3 \cdot 10^{12}} |w|^{-2m} > 0$. $\square$

By Hausdorff's theorem [18] the moment property is the complete monotonicity $\sum_{j=0}^k (-1)^j \binom{k}{j} S_{m+j+1} \geq 0$ for all $k, m \geq 0$, an explicit family of inequalities for the cumulants. The Hankel determinants of $s_m = \gamma_1^{2m} S_m$, computed from the Taylor coefficients of $\log \Xi$ by a discrete Cauchy integral, are $\det[s_{i+j+1}]_{i,j < k} = 4.62, 3.11, 0.110, 7.59 \times 10^{-5}, 6.41 \times 10^{-10}$ and $\det[s_{i+j+2}]_{i,j < k} = 1.48, 0.246, 1.26 \times 10^{-3}, 9.49 \times 10^{-8}, 8.18 \times 10^{-14}$ for $k = 1, \dots, 5$, all positive.

Proof of Theorem 1.2. Parts (a)–(f) are Theorem 3.2, Theorem 5.3, Theorem 6.1, Theorem 7.1, Theorem 7.2 and Theorem 8.5. Each is equivalent to RH, which is equivalent to Conjecture 1 by [44]. $\square$

9 The primes as generators

Theorem 9.1. *Theorem 1.8(i) holds.*

Proof. By induction, $\langle G_k \rangle$ is the set of ℓ_k -smooth integers: every $n \in [2, \ell_{k+1})$ has all its prime factors at most ℓ_k , and ℓ_{k+1} has not, so $g_{k+1} = \ell_{k+1}$. An atom of $(\mathbb{Z}_{\geq 1}, \cdot)$, an element $\neq 1$ that is not a product of two elements $\neq 1$, lies in every generating set, and every non-atom > 1 is a product of smaller elements; the atoms are the primes. In Hilbert's monoid $\{n \equiv 1 \pmod{4}\}$ the atoms 9, 21, 49 satisfy $441 = 9 \cdot 49 = 21 \cdot 21$, so the minimal generating set exists and factorisation is not unique; its Dirichlet series $\sum_{n \equiv 1(4)} n^{-s} = \frac{1}{2}[(1 - 2^{-s})\zeta(s) + L(s, \chi_{-4})]$ is a sum of two Euler products, the shape of the Davenport–Heilbronn function. Uniqueness in $\mathbb{Z}$ comes from Euclid's lemma, an additive fact. $\square$

Lemma 9.2 (the triad's digits). *Theorem 1.8(ii) holds. Moreover $n \equiv \pm 1 \pmod{6}$ if and only if n is odd and $3 \nmid n$, and then the sign is $\chi_{-3}(n)$.*

Proof. Write $n = 3^v m$, $3 \nmid m$. The balanced-ternary expansion of n is that of m shifted by v places, and the last digit $d \in \{\pm 1\}$ of m satisfies $m \equiv d \pmod{3}$, so $d = \chi_{-3}(m)$; complete multiplicativity follows, and $\sum_n d(n)n^{-s} = \sum_v 3^{-vs} \sum_{3 \nmid m} \chi_{-3}(m)m^{-s}$. Since $3^i \equiv 1 \pmod{2}$, $s_3(n) \equiv n \pmod{2}$, and $\sum (-1)^n n^{-s} = -\eta(s)$. The last statement is the Chinese remainder theorem. $\square$

With $L_{\pm}(x) = \sum_{n \leq x, n \equiv \pm 1(6)} \lambda(n)$, the Euler products give $\sum_{(n,6)=1} \lambda(n)n^{-s} = (1+2^{-s})(1+3^{-s})\zeta(2s)/\zeta(s)$ and $\sum_{(n,6)=1} \lambda(n)\chi_{-3}(n)n^{-s} = (1-2^{-s})(1-3^{-2s})\zeta(2s)/L(s, \chi_{-3})$; as for [44, Prop. 3.1], RH is equivalent to $L_+ + L_- = O(x^{1/2+\varepsilon})$, and the Riemann Hypothesis for $L(s, \chi_{-3})$ to $L_+ - L_- = O(x^{1/2+\varepsilon})$. The terms of the poles at $s = \frac{1}{2}$ are $-1.8439\sqrt{x}$ and $+0.4061\sqrt{x}$, and the logarithmic means of $(L_+ \pm L_-)/\sqrt{x}$ over $[10^4, 10^{10}]$ are -1.8383 and $+0.4077$.

Proposition 9.3 (the observer-induction barrier). *Theorem 1.8(iii) holds: for signs $\varepsilon = (\varepsilon_{\ell})$ put $M_{\varepsilon}(x, k) = \sum \mu^2(n) \prod_{\ell|n} \varepsilon_{\ell}$ over the ℓ_k -smooth $n \leq x$. Then $M_{\varepsilon}(x, k) = M_{\varepsilon}(x, k-1) + \varepsilon_{\ell_k} M_{\varepsilon}(x/\ell_k, k-1)$, the bound $B(x, k) = B(x, k-1) + B(x/\ell_k, k-1)$, $B(x, 0) = \mathbf{1}_{x \geq 1}$, that the triangle inequality gives is the number $Q_k(x)$ of squarefree ℓ_k -smooth $n \leq x$, and $\max_{\varepsilon} |M_{\varepsilon}(x, k)| = Q_k(x)$. For every $\sigma \geq 0$, $Q_k(x) \leq x^{\sigma} \prod_{i \leq k} (1 + \ell_i^{-\sigma})$, and the product is bounded in k exactly when $\sigma > 1$.*

Proof. Split the squarefree ℓ_k -smooth n by whether $\ell_k \mid n$. The recursion for B counts the same set, and $\varepsilon \equiv +1$ attains it; $M(x)$ is the case $\varepsilon \equiv -1$ with $k = \pi(x)$. Rankin's bound is $Q_k(x) \leq \sum_n (x/n)^{\sigma}$ over squarefree ℓ_k -smooth n , and $\prod (1 + \ell^{-\sigma})$ converges exactly for $\sigma > 1$. $\square$

Proposition 9.4 (the observer path). *Theorem 1.8(iv) holds. More generally, for $y \geq \sqrt{x}$, $M(x, y) = M(x) + \sum_{m \leq x/y} \mu(m)(\pi(x/m) - \pi(y))$, and for $1 < u \leq 2$, $L(x, x^{1/u}) = \sum_{n \leq x, P^+(n) \leq x^{1/u}} \lambda(n) = -\zeta(2)x/\log^2 x + O_u(x/\log^3 x)$.*

Proof. If $n \leq x$ has a prime factor $\ell > y \geq \sqrt{x}$, then $n = \ell m$ with $m \leq x/y < \ell$ and $\mu(n) = -\mu(m)$, which gives the identity, and $y = x/2$ leaves only $m = 1$. Put $z = x^{1-1/u} \leq y$ and $\mathcal{L} = \log x$. By the prime number theorem with the classical error term [55, Ch. III], $M(t), L(t) \ll t e^{-c\sqrt{\log t}}$ and $\pi(t) = \text{li}(t) + O(te^{-c\sqrt{\log t}})$; since $\log(x/m) \geq \mathcal{L}/u$ for $m \leq z$, the errors, $M(x)$ and $\pi(y)M(z)$ contribute $O(xe^{-c'\sqrt{\mathcal{L}}})$. Write $\text{li}(x/m) = \frac{x}{m}[(\mathcal{L} - \log m)^{-1} + (\mathcal{L} - \log m)^{-2} + O(\mathcal{L}^{-3})]$ and expand in powers of $\log m/\mathcal{L}$ up to order 3; the remainders are controlled by partial summation against $M(t)$, and the sums $\sum_{m \leq z} \mu(m)(\log m)^j/m$ equal 0, -1 , $-2\gamma_E$ for $j = 0, 1, 2$ up to $O(e^{-c\sqrt{\log z}})$, from $1/\zeta(1+w) = w - \gamma_E w^2 + O(w^3)$. Collecting

terms, $\sum_{m \leq z} \mu(m) \operatorname{li}(x/m) = x[-\mathcal{L}^{-2} - 2\gamma_E \mathcal{L}^{-3} - 2\mathcal{L}^{-3}] + O(x\mathcal{L}^{-4})$. For L the same argument with $\zeta(2+2w)/\zeta(1+w) = \zeta(2)w + O(w^2)$ gives the leading term. $\square$

The record to 10^{10} . An exact segmented sieve over all $n \leq 10^{10}$, in integer arithmetic, gives $\pi(10^{10}) = 455,052,511$, $M(10^{10}) = -33,722$ and $L(10^{10}) = -116,026$, in agreement with the published values [34, 35, 36], and the first $x > 1$ with $L(x) > 0$ at 906,150,257 [54]. $L(x) > 0$ at 305,426 integers $x \leq 10^{10}$ and M changes sign 131,477 times. On $[10^4, 10^{10}]$: $(\psi(x) - x)/\sqrt{x} \in [-0.751, 0.791]$; $(\pi(x) - \operatorname{li}(x)) \log x/\sqrt{x} \in [-2.447, -0.421]$; $M(x)/\sqrt{x} \in [-0.464, 0.571]$; $L(x)/\sqrt{x} \in [-1.3583, 0.0275]$. Schoenfeld's bound $|\psi(x) - x| < \sqrt{x} \log^2 x/8\pi$ for $x \geq 73.2$, which holds under RH [42], is met with ratio at most 0.918 (as $x \uparrow 97$), and at most 0.0396 on $[10^9, 10^{10}]$. At $x = 10^{10}$ the observer path gives $M(x, \sqrt{x}) = -21,940,269 = -219.40\sqrt{x}$, against $-214.45\sqrt{x}$ from Proposition 9.4; $L(x, \sqrt{x}) = -373.17\sqrt{x}$; $M(x, x^{2/3}) = -212.98\sqrt{x}$; $M(x, x/2) = 220,064,566 = 2200.6\sqrt{x}$; and $M(x) = -0.337\sqrt{x}$. The balance at the scale $\sqrt{x}$ is reached only when the last primes, those in $(x/2, x]$, are added, and their number is $\pi(x) - \pi(x/2)$.

Beurling systems. Generalised prime systems in the sense of Beurling are generated freely by their primes, so Theorem 1.8(i) holds for them by construction. Hilberdink [19] proved that if $N_B(x) = ax + O(x^{\beta+\varepsilon})$ and $\psi_B(x) = x + O(x^{\alpha+\varepsilon})$ then $\max(\alpha, \beta) \geq \frac{1}{2}$, and Broucke, Debruyne and Révész [6] constructed, for every $\alpha \in [0, 1)$ and $\beta \in [\frac{1}{2}, 1)$, a system for which these are the optimal exponents. So the generating rule, together with the regularity of the integers at the square-root scale, does not force $\alpha = \frac{1}{2}$; for $\mathbb{Z}$ the statement $\alpha = \frac{1}{2}$ is RH.

10 The comparison functions

Proposition 10.1 (shared structure). *Let $F \in \{\Xi, \Xi_\epsilon, \Xi_{\text{DH}}\}$. Then F is even, entire of order one, real on $\mathbb{R}$, with $F(s) = F(1-s)$ in the variable $s = \frac{1}{2} + iz$, and its zeros lie in a horizontal strip. For each such F : the pair formula (Proposition 2.1); Theorem 3.1; the Jensen–Riesz formula of Theorem 3.2 with Y above the strip; the identities of Proposition 5.2, the index formula of Theorem 5.3 and the count of Corollary 5.5 (whose inclusion in $\mathcal{U}$ is specific to ζ); the shadow law (Theorem 5.6); and the Hermite and landing laws of Theorems 6.2 and 6.3 for its heat flow. For $F \in \{\Xi, \Xi_\epsilon\}$, which are transforms of positive even doubly-exponentially decaying kernels, also the Husimi inequality (Theorem 5.8) and the Macdonald expansion with tail positivity of the Wigner function [44]. Ξ_ϵ and Ξ_{DH} have zeros off the real axis. Hence no conjunction of these properties implies that the zeros are real.*

Proof. The proofs cited use only the stated properties; the strip for Ξ_ϵ is Proposition 10.3(ii), and for Ξ_{DH} , $|f(s) - 1| \leq \sum_{n \geq 2} n^{-\sigma} < 1$ for $\sigma \geq \frac{5}{2}$, so its zeros have $|\operatorname{Im} z| < 2$. The off-line zeros are s_0 and Spira's zeros [44, 53]. $\square$

Proposition 10.2 (shared inequalities, computed). *The following hold for ζ (the cited theorems) and, as computed, for Z_ϵ .*

- (i) *The Turán inequalities $b_k^2 > \frac{2k-1}{2k+1} b_{k-1} b_{k+1}$ for the moments $b_k = \int_0^\infty \Phi(u) u^{2k} du$, proved for all k for Φ by Csordas, Norfolk and Varga [10]; for Φ_ϵ for $k \leq 69$, with minimum margin $1 - \frac{2k-1}{2k+1} b_{k-1} b_{k+1} / b_k^2$ of 7.37×10^{-3} (7.03×10^{-3} for Φ).*
- (ii) *Newman's convexity $V''' > 0$ for $V = -\log \Phi$ on $[0, \infty)$ [32]; for Φ_ϵ , $\min V'''(u)/u = 6.871$ on $[0.02, 2.5]$ (129.08 for Φ).*
- (iii) *$(\log Z)'''(h) \leq 0$ for $Z(h) = \frac{1}{2} \int K(u) e^{hu} du$: maximum -4.46×10^{-6} on $(0, 80]$ for Φ , -2.17×10^{-4} on $(0, 40]$ for Φ_ϵ .*

- (iv) The signs of the Lee–Yang cumulants, $S_m > 0$: for all m for ζ (Theorem 8.5), and for $m \leq 30$ for Z_ϵ .
- (v) $\lambda_n^\epsilon > 0$ for $n \leq 1129$.
- (vi) $\mu(V_{0.1}(L)) > 0$ for $L \leq 1.6$ (the Weil form on low-frequency windows).

Proposition 10.3 (separating properties, proved). (i) ζ has an Euler product; Z_ϵ and f have none ($r_\epsilon(1) = 0$, and $a(4) - a(2)^2 = -1.0807$ [44]).

- (ii) $\zeta(s) \neq 0$ for $\operatorname{Re} s \geq 1$, while Z_ϵ has infinitely many zeros with $\operatorname{Re} s > 1$ [12] and none with $\operatorname{Re} s \geq 2.1$. Hence $E_{1/2}$ is of Hermite–Biehler class for Ξ and not for Ξ_ϵ (Proposition 5.1), and $\Lambda_\epsilon \leq 5.12$.
- (iii) $R_\epsilon \leq |1 - 1/s_0| = 0.992700$, while $R_\zeta^2 \geq 1 - 1/(9 \cdot 10^{24})$ and $R_\zeta = 1$ under RH.
- (iv) f has infinitely many zeros with $\frac{1}{2} < \operatorname{Re} s < 1$, by the theorem of Kaczorowski and Kulas on combinations $\sum_j P_j L(s, \chi_j)$ of L -functions of nonequivalent characters [20]; so $N_+^{\text{DH}} = \infty$, and likewise $N_+^\epsilon = \infty$ by (ii).
- (v) Explicitly $N_+^\epsilon \geq 14$ below height 80: Z_ϵ has the zeros $1.159587 + 9.452251i$, $0.679074 + 43.158494i$, $1.228628 + 52.726152i$, $1.021329 + 57.072384i$, $0.844026 + 65.210123i$, $1.527823 + 71.427717i$, $0.882604 + 79.175290i$ with $\beta > \frac{1}{2}$, each giving two zeros of Ξ_ϵ in $\mathbb{C}_+$.

Proof. (ii) For $\sigma \geq 2.1$, $|Z_\epsilon(s)| \geq 2 \cdot 2^{-\sigma} - \sum_{n \geq 3} r_\epsilon(n) n^{-\sigma}$, and $\sum_{n \geq 3} r_\epsilon(n) (n/2)^{-\sigma}$ decreases in σ . At $\sigma = 2.1$ the exact values of $r_\epsilon(n)$ for $n \leq 4 \cdot 10^5$ give $\sum_{3 \leq n \leq 4 \cdot 10^5} r_\epsilon(n) n^{-2.1} = 0.4664970$, against $2 \cdot 2^{-2.1} = 0.4665165$, and the bound $\#\{(m, k) : q_\epsilon(m, k) \leq t\} \leq (\sqrt{2t} + 1)(2\sqrt{t/3} + 1)$ bounds the rest by 1.3×10^{-6} . So the zeros of Ξ_ϵ have $|\operatorname{Im} z| < 1.6$, and de Bruijn’s theorem (Section 6) gives real zeros for $T \geq 2 \cdot 1.6^2 = 5.12$. (iii) $1 - 1/s_0$ is a singularity of the Li generating function of ξ_ϵ (Theorem 7.2). (iv) $f = \frac{1}{2} \sec \vartheta [e^{-i\vartheta} L(s, \chi) + e^{i\vartheta} L(s, \bar{\chi})]$ with $\chi \neq \bar{\chi}$ [44]. (v) Each zero was located by Newton’s method on ξ_ϵ , with final Newton correction below 1.5×10^{-14} , and each has its mirror zero at $1 - \bar{s}$. $\square$

Proposition 10.4 (separating properties, computed). (i) $\Lambda_\epsilon \geq 0.7472$: the first off-line pair of Ξ_ϵ reaches the real axis at $T_\epsilon = 0.747230705896$, and $\min \mathcal{L}_T = -1.47 \times 10^{-12}$ at $T = 0.7472$ against a rounding level of 5×10^{-16} . For ζ , $\Lambda \leq 0.2$.

- (ii) $\lambda_{1130}^\epsilon = -414.9$ after $\lambda_{1129}^\epsilon = 74.2$, with 192 negative coefficients for $n \leq 1600$, from the growth 1.00735^n of the contribution of the mirror zero $1 - \bar{s}_0$; for ζ , $\lambda_n > 0$ for $n \leq 10^{12}$ (Theorem 1.4).
- (iii) The Weil form of Z_ϵ is negative on low-frequency windows: $\mu(V_h(L))$ changes sign at $L = 1.741, 1.653, 1.610, 1.587$ for $h = 0.1, 0.05, 0.025, 0.0125$ (extrapolated threshold $L^* \approx 1.56$), and $\mu(V_{0.1}(L)) = -0.0523, -1.42, -6.47, -99.5, -388$ at $L = 2, 2.5, 3, 5, 6$. For f near height $T_0 = 85.699348$, $\mu(V_h(L, T_0)) < 0$ for $L \geq 3.916$ ($h = 0.1$) and $L \geq 3.812$ ($h = 0.05$), with $\mu = -0.181, -1.61, -3.27$ at $L = 4, 5, 6$, of which the quadruple of Spira’s zero carries -0.805 at $L = 4$. On the same spaces ζ has $\mu(V_{0.1}(L)) > 0$ for $L \leq 5$ and $\mu(V_{0.1}(L, T_0)) = 1.11, 5.1 \times 10^{-3}, 1.35 \times 10^{-8}, 2.6 \times 10^{-10}, 1.5 \times 10^{-12}$ at $L = 2, 3, 4, 5, 6$.
- (iv) The Hankel determinant $\det[s_{i+j+2}^\epsilon]_{i,j < 4}$ of $s_m^\epsilon = 5.1308^{2m} S_m^\epsilon$ is -9.88×10^{-10} , and $\det[s_{i+j+1}^\epsilon]_{i,j < 5} = -5.40 \times 10^{-12}$; so (S_{m+1}^ϵ) is not a moment sequence, as Theorem 8.5 requires. For ζ the corresponding determinants are 9.49×10^{-8} and 6.41×10^{-10} .
- (v) The prime side of Z_ϵ has coefficients of both signs: with $-\frac{d}{ds} \log(Z_\epsilon(s)/2^{1-s}) = \sum_Q c_Q Q^{-s}$ over products Q of the numbers $n/2$, $c_Q = +0.405, -0.405, +1.833, +0.405, -2.644, +1.386$ at $Q = \frac{3}{2}, \frac{9}{4}, \frac{5}{2}, \frac{27}{8}, \frac{15}{4}, 4$; for ζ the coefficients $\Lambda(n)$ are nonnegative. For f , $-f'/f$ has coefficients $1.936, -0.763, -2.852$ at $n = 6, 12, 14$, which are not prime powers.

At $x_0 = 9.45225$ the ledger of Z_ϵ has $G_\epsilon \approx 2.01$ and $A_\epsilon = 3.62, 7.21, 10.09, 105.3, 1703$ at $y = 0.3, 0.5, 0.55, 0.65, 0.659$; the prime side exceeds the archimedean side below the zero and changes sign at $y = 0.6596$, the height of the zero. Table 3 collects the separating properties and Fig. 2 shows two of them.

property	ζ	Z_ϵ	Davenport–Heilbronn f
Euler product	yes	no	no
zeros with $\text{Re } s \geq 1$	none	infinitely many	—
de Bruijn–Newman constant	$[0, 0.2]$	$[0.7472, 5.12]$	—
Li radius	$\geq 1 - 6 \times 10^{-26}$; 1 under RH	≤ 0.9927	—
first negative Li coefficient	none for $n \leq 10^{12}$	λ_{1130}	—
Weil form negative for window length	no $L \leq 5$ (computed)	$L \geq 1.587$	$L \geq 3.812$ (at T_0)
index N_+	0 below $3 \cdot 10^{12}$	∞ (14 below 80)	∞ (8 below 200)
prime-side coefficients	$\Lambda(n) \geq 0$	both signs	both signs

Table 3: Properties that separate ζ from Z_ϵ and from the Davenport–Heilbronn function. The counts below 80 and 200 come from the zeros listed in Proposition 10.3 and from Spira’s four zeros.

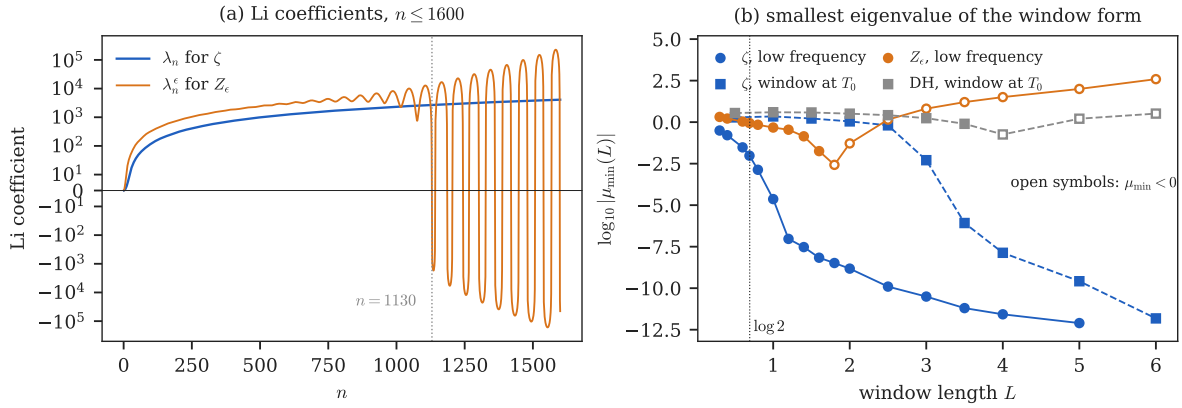


Figure 2: (a) Li coefficients of ζ and of Z_ϵ for $n \leq 1600$ (symmetric logarithmic scale). Those of Z_ϵ oscillate with growth 1.00735^n from the mirror of the zero s_0 and are first negative at $n = 1130$. (b) The smallest eigenvalue μ of Weil’s form on spline spaces of window length L (knot spacing 0.1): ζ and Z_ϵ at low frequency, ζ and the Davenport–Heilbronn function near height $T_0 = 85.699$. Filled symbols are positive values and open symbols negative ones; the dotted line is $L = \log 2$.

11 Methods of computation

Arithmetic. All computations except the sieve use IEEE double precision (about 16 significant digits), adaptive Gauss–Kronrod quadrature with relative tolerance 10^{-10} to 10^{-13} , and the trapezoidal rule for integrands that are analytic in a strip, which converges geometrically. The sieve to 10^{10} is exact integer arithmetic, segmented in blocks, with ψ and θ accumulated in double precision.

The functions. $\zeta(s)$, $\zeta'(s)$ and the Hurwitz function $\zeta(s, \alpha)$ with its derivative are computed by Euler–Maclaurin summation with $\max(60, |s|/\pi + 40)$ terms and 20 to 30 Bernoulli corrections, the Bernoulli numbers exact as rationals. Dirichlet L -functions are combinations of Hurwitz values; Z_ϵ is computed from the genus identity for $\text{Re } s < 4$ and from its Dirichlet series with the exact $r_\epsilon(n)$, $n \leq 2 \times 10^5$, for $\text{Re } s \geq 4$, and agrees with the lattice sum over $|m|, |n| \leq 400$ at $s = 2.5$ and 3. The functional equations of ξ , ξ_ϵ and F_{DH} hold to 7.5×10^{-15} relative at $s = 0.3 + 7.7i$; $\zeta(2) - \pi^2/6 = 2 \times 10^{-16}$; and $v(x, 0) = 0$ off the zeros holds to 10^{-13} for $x \leq 120$, 2×10^{-8} at $x = 7005$ and 2×10^{-7} at $x = 17143$. The Gamma and digamma functions at complex argument are evaluated by recurrence and the Stirling series. Ξ_T , Ξ_ϵ and their derivatives are

computed from the kernel integrals by the trapezoidal rule on $[0, 3.5]$ and $[0, 8]$, as in [44].

Zeros. The zeros of ζ on the line below 3000 (2469 of them) are located by Brent's method on sign changes of Hardy's Z on a grid of $1/12$ of the mean spacing, and refined by Newton's method on ξ ; $\gamma_1 = 14.134725141734696$ agrees with Odlyzko's table [33] to 3×10^{-15} . The zeros of ξ_ϵ , F_{DH} , $\Xi + it\Xi'$ and of the heat-flowed functions are located by Newton's method from grid minima, and counted by the argument principle on rectangles.

The flow and the surgery. v and u are computed from ξ'/ξ . The sets $\{v < 0\}$ are found on rectangular grids of 3×10^{-3} to 10^{-2} in each direction. The Poisson integrals of Theorem 3.2 are computed by adaptive quadrature on $|t - x_0| \leq 12$ with the Green sums over the zeros in that range. The surgery quantities use the closed form of R .

Li coefficients and Weil windows. The coefficients of $F'(w) = \xi'/\xi(1/(1-w))(1-w)^{-2}$ are computed by the discrete Cauchy integral on $|w| = r$ with M equally spaced nodes (fast Fourier transform), $(r, M) = (0.99, 8192)$ and $(0.9998, 2^{18})$ for ζ and $(0.975, 16384)$, $(0.99, 16384)$ for ξ_ϵ , whose coefficients agree to 0.07 at $n = 1130$; the imaginary parts, zero in exact arithmetic, are below 8.2×10^{-7} for ζ ($n \leq 30000$) and 8.1×10^{-9} for ξ_ϵ ($n \leq 1200$, $r = 0.99$). The Weil form is computed from the contour integral of Section 7 by the trapezoidal rule with step 0.1 on $0 \leq t \leq 1500$, at $c = 1.5$ and 2.6 for ζ and $c = 2.6$ and 3.2 for ξ_ϵ and F_{DH} (beyond all their zeros), and $\mu(V)$ from the generalised eigenvalue problem with the L^2 Gram matrix of the basis. The Lee–Yang sums S_m are computed from the Taylor coefficients of $\log \Xi(-ih)$ by a discrete Cauchy integral with 512 nodes on $|h| = 10$ (ζ) and $|h| = 4$ (ξ_ϵ), inside the first zeros 14.13 and 5.131.

The heat flow and the spins. Ξ_T and its derivatives are computed from the kernel integral; the landing time is found by following the zero by Newton's method in T ; the backward collision times of the Lehmer pairs by locating the change of sign of the local extremum of Ξ_T between the two zeros. On $|z| = 1$, $Z_{n,T}(e^{i\vartheta}) = P(\cos \vartheta)$ with P a real polynomial of degree n , so all $2n$ zeros lie on the circle exactly when P has n zeros in $[-1, 1]$; these are counted by Sturm sequences in 80-digit decimal arithmetic, and $\Lambda_n(K)$ is located by bisection in T after a scan of step 10^{-3} downward from $T = 0$.

12 Concluding remarks

We have shown that the sizes of $\log |\Xi|$ and its gradient on $\text{Re } s \geq 1$, known to a tolerance $C_1/\log^2(|x| + 2)$, together with the verified zeros, the zero-free region and the zero count, do not decide the sign of the current in $\mathcal{U}$, and we have given six equivalent forms of Conjecture 1: a single-point Jensen–Riesz defect, the Krein–Langer index of the current, the absence of multiple real zeros of the heat flow on $(0, 0.2]$, Weil positivity on the Poisson family, the Li radius and a Hausdorff moment property. Unconditionally, the downdraft has measure $\ll T/\log T$ in $[T, 2T]$ and touches the zeros off the line, Ford's region carries the updraft, and $\lambda_n > 0$ for $n \leq 10^{12}$. The Epstein and Davenport–Heilbronn functions satisfy the structural statements of Proposition 10.1 and differ from ζ in the Euler product and in the properties of Table 3. Conjecture 1 is the remaining step; by Theorem 1.1 a proof of it must use the arithmetic of ζ exactly, not through estimates on $\text{Re } s \geq 1$, and the comparison functions single out the Euler product as the arithmetic property that separates ζ from them.

Outlook. The Euler factors are Lee–Yang polynomials in ℓ^{-s} with their zeros on $\operatorname{Re} s = 0$, and the next step is to transport this positivity across the strip, through the merge of copies or the heat flow, in a form that the Epstein and Davenport–Heilbronn functions fail. The index formula of Theorem 1.5 and the window forms of Section 7 give finite-dimensional quantities on which such a transport can be measured.

13 Data and sources

External inputs. All inputs are mathematical; no measured data are used.

- (i) All nontrivial zeros of ζ with $0 < \gamma \leq 3 \cdot 10^{12}$ lie on the critical line, and $\Lambda \leq 0.2$ [37]; $\Lambda \geq 0$ [41].
- (ii) Zero-free regions: $\sigma \geq 1 - 1/(5.573412 \log |t|)$, $|t| \geq 2$ [29]; $1 - \sigma \leq 1/(57.54(\log |t|)^{2/3}(\log \log |t|)^{1/3})$, $|t| \geq 3$ [16, Thm. 5].
- (iii) $|N(T) - \frac{T}{2\pi} \log \frac{T}{2\pi e} - \frac{7}{8}| \leq 0.112 \log T + 0.278 \log \log T + 2.510 + 0.2/T$ for $T \geq e$ [56].
- (iv) Selberg’s density theorem $N(\sigma, T) \ll T^{1-(\sigma-1/2)/4} \log T$ [43] and Selberg’s positive proportion of zeros on the line in intervals [55, Ch. X].
- (v) The uniform bound on the real parts of the non-real zeros of the heat-flowed function, Theorem 1.5(i) of [39], and the simplicity of the zeros for $T > \Lambda$ [11, Cor. 1].
- (vi) Schoenfeld’s bound $|\psi(x) - x| < \sqrt{x} \log^2 x / 8\pi$, $x \geq 73.2$, under RH [42].
- (vii) Published values used for comparison: $\pi(10^{10}) = 455,052,511$ [34]; $M(10^{10}) = -33,722$ [35]; $L(10^{10}) = -116,026$ [36]; the least $x > 1$ with $L(x) > 0$, 906,150,257 [54]; γ_1 [33]; Spira’s zeros of the Davenport–Heilbronn function [53]; the values -5.389×10^{-69} and -1.31957×10^{-4} of Conrey and Li [9].
- (viii) The zero s_0 of Z_ϵ , the kernels Φ and Φ_ϵ and the Macdonald expansion are from [44].

Computed quantities. Every other number in the paper is computed from formulas stated in the text by the methods of Section 11: the zeros of ζ below 3000, the Lehmer pairs and the zeros of ξ_ϵ and of $\Xi + it\Xi'$; the flow fields, the Jensen–Riesz defects and the surgery quantities; the Li coefficients of ζ and Z_ϵ ; the Weil window eigenvalues; the ν -flow velocities; the truncation zeros; the de Branges statistics; the heat-flow landing and collision times; the spin-chain constants; the Lee–Yang sums and Hankel determinants; the Turán margins, V''' and $(\log Z)'''$; $r_\epsilon(n)$ for $n \leq 4 \cdot 10^5$; and the sieve data to 10^{10} .

Contributions and use of AI. The ideas, the research questions, the framework that links the fields, the choice of problems and the direction at every stage are the author’s. The derivations, proofs, numerical checks, reference checks and drafting were produced with an AI system working under that direction, and were checked by separate AI review passes. Errors in that material are errors of the AI system, and any found will be corrected in a revised version. Publication-ethics policy (COPE, *Authorship and AI tools*, 2023) does not allow an AI system to be listed as an author, so its use is declared here instead.

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